

Aligning Cyber Space with Physical World: A Comprehensive Survey on Embodied AI

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Abstract—Embodied Artificial Intelligence (Embodied AI) is crucial for achieving Artificial General Intelligence (AGI) and serves as a foundation for various applications (e.g., intelligent mechatronics systems, smart manufacturing) that bridge cyberspace and the physical world. Recently, the emergence of Multi-modal Large Models (MLMs) and World Models (WMs) have attracted significant attention due to their remarkable perception, interaction, and reasoning capabilities, making them a promising architecture for embodied agents. In this survey, we give a comprehensive exploration of the latest advancements in Embodied AI. Our analysis firstly navigates through the forefront of representative works of embodied robots and simulators, to fully understand the research focuses and their limitations. Then, we analyze four main research targets: 1) embodied perception, 2) embodied interaction, 3) embodied agent, and 4) sim-to-real adaptation, covering state-of-the-art methods, essential paradigms, and comprehensive datasets. Additionally, we explore the complexities of MLMs in virtual and real embodied agents, highlighting their significance in facilitating interactions in digital and physical environments. Finally, we summarize the challenges and limitations of embodied AI and discuss potential future directions. We hope this survey will serve as a foundational reference for the research community. The associated project can be found at https://github.com/HCLab-SYSU/Embodied_AI_Paper_List.

Index Terms—Embodied AI, Cyber Space, Physical World, Multi-modal Large Models, Agents, Mechatronic Intelligence

I. INTRODUCTION

EMBODIED AI was initially proposed from the Embodied Turing Test by Alan Turing in 1950 [1] and has wide

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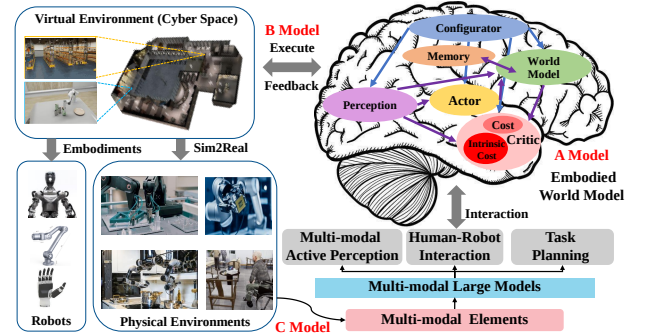


Fig. 1. The framework of the embodied agent based on MLMs and WMs, incorporates the ABC model, which stands for AI brain, Body, and Cross-modal sensors. The embodied agent is equipped with an embodied world model as the A model, enabling it to understand the virtual-physical environment. Through the C model, it actively perceives multi-modal elements, enhancing its situational awareness. Meanwhile, the B model endows the agent execute actions, and interact with humans while utilizing tools effectively.

applications including robotics, healthcare, and smart manufacturing. Embodied AI is designed to determine whether agents can display intelligence that is not just limited to solving abstract problems in a virtual environment (cyber space¹), but that is also capable of navigating the complexity and unpredictability of the physical world. For example, embodied AI enhances mechatronic systems by integrating with physical components for adaptive, real-world interactions, which enables systems to learn, perceive, and execute autonomously, improving efficiency and functionality, as shown in Fig. 1. The agents in the cyber space are generally referred to as disembodied AI, while those in the physical space are embodied AI (Table I). Recent advances in Multi-modal Large Models (MLMs) have injected strong perception, interaction and planning capabilities to embodied models, to develop general-purpose embodied agents and robots that actively interact with virtual and physical environments. Therefore, the embodied agents are widely considered as the best carriers for MLMs. The recent representative embodied models are RT-2 [2] and RT-H [3]. Nevertheless, the capabilities of long-term memory, understanding complex intentions, and the decomposition of complex tasks are limited for current MLMs.

To achieve Artificial General Intelligence (AGI), the development of embodied AI stands as a fundamental avenue. Different from conversational agents like ChatGPT [4], embodied AI believes that the true AGI can be achieved by controlling physical embodiments and interacting with both

¹The agents are the foundation of both disembodied and embodied AI. The agents can exist in both cyber and physical spaces, integrated with various entities. The entities include not only robots but also other devices.

网络空间与物理世界的对齐：具身人工智能的全面综述

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摘要——具身人工智能 (Embodied AI) 对于实现通用人工智能 (AGI) 至关重要, 并为连接网络空间与物理世界的各类应用 (例如智能机电系统、智能制造) 奠定了基础。近年来, 多模态大型模型 (MLMs) 与世界模型 (WMs) 因其卓越的感知、交互及推理能力而备受关注, 使其成为具身智能体的理想架构。本文系统综述了具身人工智能领域的最新进展: 首先梳理了代表性机器人与模拟器的研究成果, 全面解析其研究重点与局限性; 随后深入探讨四大核心研究方向——1) 具身感知; 2) 具身交互; 3) 具身智能体; 4) 仿真-现实适应机制, 涵盖前沿方法、关键范式及综合性数据集。此外, 我们深入探讨了虚拟与现实具身智能体中多层次营销 (MLM) 的复杂性, 着重分析其在数字与物理环境中促进交互作用的关键作用。最后, 我们总结了具身人工智能面临的挑战与局限性, 并探讨了潜在的发展方向。我们希望本综述能为研究领域提供基础性参考。相关项目成果可访问 <https://github.com/HCLab-SYSU/Embodied-AI-Paper-List> 获取。

索引术语——具身人工智能、网络空间、物理世界、多模态大型模型、智能体、机电智能

I. 引言

具身人工智能最初由艾伦·图灵于1950年基于具身图灵测试提出[1]并具有广泛应用价值。本研究部分获得以下资助: 国家重点研发计划项目 (项目编号2021ZD0111601); 鹏程实验室开放研究基金 (项目编号2025KF1B0050); 国家自然科学基金项目 (项目编号62436009、62322608及62301532); 以及广东省基础与应用基础研究基金 (项目编号2025A1515011874和2023A1515011530)。(通讯作者: 梁林。)

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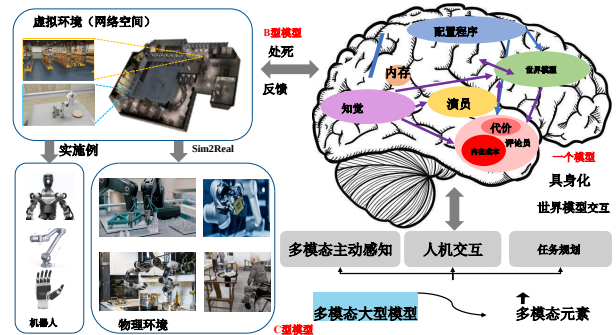


图1. 基于MLMs和WMs的具身智能体框架整合了ABC模型, 该模型代表AI大脑、身体及跨模态传感器。该具身智能体配备具身世界模型作为A模型, 使其能够理解虚拟-物理环境; 通过C模型, 它主动感知多模态元素, 提升情境感知能力; 而B模型则使智能体能够执行动作、与人类互动并有效使用工具。

应用领域涵盖机器人技术、医疗保健及智能制造等领域。具身人工智能旨在评估智能体是否具备超越虚拟环境 (网络空间¹) 中抽象问题解决能力的智能表现, 同时能够应对物理世界中的复杂性和不可预测性。例如, 具身人工智能通过与物理组件集成实现自适应现实交互, 从而增强机电系统的学习能力、感知能力和自主执行能力, 显著提升系统效率与功能表现 (如图1所示)。网络空间中的智能体通常称为非具身人工智能, 而物理空间中的则属于具身人工智能 (表I)。多模态大型模型 (MLMs) 的最新进展为具身模型注入了强大的感知、交互与规划能力, 推动了能主动与虚拟及物理环境交互的通用型具身智能体与机器人研发。因此, 具身智能体被广泛视为MLMs的最佳载体。近期具有代表性的具身模型包括RT-2[2]和RT-H[3]。然而, 当前多层模型 (MLMs) 在长期记忆能力、理解复杂意图以及分解复杂任务方面的能力仍存在局限。

要实现通用人工智能 (AGI), 具身人工智能的发展成为关键路径。与ChatGPT等对话式智能体[4]不同, 具身人工智能认为, 真正的AGI可通过控制物理具身形态并与其交互来达成。

¹智能体是无实体与有实体人工智能的基础。智能体可存在于网络空间与物理空间中, 并与各类实体集成。这些实体不仅包括机器人, 还包括其他设备。

TABLE I
COMPARISON BETWEEN DISEMBODED AI AND EMBODIED AI.

Type	Environment	Physical Entities	Description	Representative Agents
Disembodied AI	Cyber Space	No	Cognition and physical entities are disentangled	ChatGPT [4], RoboGPT [8]
Embodied AI	Physical Space	Robots, Cars, Other devices	Cognition is integrated into physical entities	RT-1 [9], RT-2 [10], RT-H [3]

simulated and physical environments [5], [6]. As we stand at the forefront of AGI-driven innovation, it is crucial to delve deeper into the realm of embodied AI, unraveling their complexities, evaluating their current developmental stage, and contemplating the potential trajectories they may follow in the future. Nowadays, embodied AI contains various key techniques across Computer Vision (CV), Natural Language Processing (NLP), and robotics, with the most representative being embodied perception, embodied interaction, embodied agents, and sim-to-real robotic control [7]. Therefore, it is imperative to capture the evolving landscape of embodied AI in the pursuit of AGI through a comprehensive survey.

Embodied agent is the most prominent basis of embodied AI. For an embodied task, the embodied agent must fully understand the human intention in language instructions, actively explore the surrounding environments, comprehensively perceive the multi-modal elements from both virtual and physical environments, and execute appropriate actions for complex tasks [11], [12], as shown in Fig. 1. The rapid progress in multi-modal models exhibits superior versatility, dexterity, and generalizability in complex environments compared to traditional deep reinforcement learning approaches. Pre-trained visual representations from state-of-the-art vision encoders [13], [14] provide precise estimations of object class, pose, and geometry, which makes the embodied models thoroughly perceive complex and dynamic environments. Powerful Large Language Models (LLMs) make robots better understand the linguistic instructions from humans. Promising MLMs give a feasible approach for aligning the visual and linguistic representations of embodied robots. The world models [15], [16] exhibit remarkable simulation capabilities and promising comprehension of physical laws, which makes embodied models comprehensively understand both the physical and real environments. These innovations empower embodied agents to comprehensively perceive complex environment, interact with humans naturally, and execute tasks reliably.

Despite the intensive interest in harvesting the powerful perception and reasoning ability from MLMs, the research community is short of a comprehensive survey that can help sort out existing embodied AI studies, the challenges faced, as well as future research directions. In the era of MLMs, we aim to fill up this gap by performing a systematic survey of embodied AI across cyber space to physical world. We conduct the survey from different perspectives including embodied robots, simulators, four representative embodied tasks (visual active perception, embodied interaction, multi-modal agents and sim-to-real adaptation), and future research directions. We believe that this survey will provide a clear big picture of what we have achieved, and we could further achieve along this emerging yet very prospective research direction.

Differences from previous works: Although there have been several survey papers [5], [6], [17], [18] for embodied AI, most of them are outdated as they were published before the

era of MLMs, which started around 2023. To the best of our knowledge, there are only two survey papers [6], [18] after 2023, which focused on vision-language-action models and embodied AI system for smart manufacturing, respectively. Embodied AI, with AI brain, Body and Cross-modal sensors, is first proposed in [18], which is also the first work to propose technical architecture of embodied AI system for future smart manufacturing in the era of foundation model. However, the MLMs, WMs and embodied agents are not fully considered in previous surveys. Additionally, recent developments in embodied robots and simulators are also overlooked. To address the scarcity of comprehensive survey papers in this rapidly developing field, we propose this comprehensive survey that covers representative embodied robots, simulators, and four main research tasks: embodied perception, embodied interaction, embodied agents, and sim-to-real adaptation. In summary, the main contributions of this work are threefold:

- To the best of our knowledge, this is the first comprehensive survey of embodied AI from the perspective of the alignment of cyber and physical spaces based on MLMs and WMs, offering novel insights about methodologies, benchmarks, challenges, and applications.
- We categorize and summarize embodied AI into several essential parts including robots, simulators, and four main research tasks: embodied perception, embodied interaction, embodied agents and sim-to-real adaptation, which serve as a detailed taxonomy of embodied AI.
- To facilitate the development of robust, general-purpose embodied agents, we propose a new dataset standard ARIo (All Robots In One) and a unified large-scale ARIo dataset, encompassing approximately 3 million episodes collected from 258 series and 321,064 tasks.

The rest of this survey is organized as follows. Section 2 introduces embodied robots. Section 3 describes general and real-scene embodied simulators. Section 4 introduces embodied perception, including active visual perception and visual language navigation. Section 5 introduces embodied interaction. Section 6 introduces embodied agents including the embodied multi-modal foundation model and embodied task planning. Section 7 introduces sim-to-real adaptation including embodied world model, data collection and training. In Section 8, we discuss promising research directions.

II. EMBODIED ROBOTS

Embodied agents interact with the physical environment, including robots, smart appliances, and autonomous vehicles, etc. Fixed-base robots, shown in Fig. 2 (a), are used in laboratory automation and industry due to their precision, e.g., Franka Emika panda [19], [20], Kuka iiwa [21], [22], and Sawyer [23], [24]. Wheeled robots, depicted in Fig. 2 (b), are efficient in logistics and warehousing due to their simple structure and low cost, like Kiva and Jackal robots [25]. They face challenges on uneven terrain. Tracked robots, shown in

表I
无实体人工智能与有实体人工智能的比较。

类型	环境	物理实体	描述	代表性代理人
无实体AI 具身AI	网络空间 物理空间	不 机器人、汽车及其他设备	认知与物理实体是相互独立的；认知被整合到物理实体之中。	ChatGPT [4], RoboGPT [8] RT-1 [9], RT-2 [10], RT-H [3]

模拟环境与物理环境[5]、[6]。作为人工智能驱动创新的前沿阵地，我们亟需深入探索具身人工智能领域，解析其复杂性、评估当前发展阶段，并展望其未来可能的发展路径。当前，具身人工智能融合了计算机视觉（CV）、自然语言处理（NLP）与机器人技术等关键领域，其中最具代表性的是具身感知、具身交互、具身智能体以及仿真到真实机器人控制[7]。因此，通过全面综述来把握具身人工智能在追求通用人工智能过程中的演进态势至关重要。

具身智能体是具身人工智能最核心的基础架构。对于具身任务而言，该智能体必须全面理解人类语言指令中的意图、主动探索周围环境、综合感知虚拟与物理环境中的多模态元素，并针对复杂任务执行恰当动作[11]、[12]（如图1所示）。相较于传统深度强化学习方法，多模态模型的快速发展展现出在复杂环境中更卓越的通用性、灵活性与泛化能力。来自前沿视觉编码器的预训练视觉表征[13]、[14]能够精准估计物体类别、姿态及几何特征，使具身模型能全面感知复杂动态环境。强大的大语言模型（LLMs）可提升机器人对人类语言指令的理解能力。前景广阔的MLM技术为具身机器人视觉表征与语言表征的对齐提供了可行方案。世界模型[15]、[16]展现出卓越的仿真能力与对物理定律的深刻理解，使具身模型能够全面把握物理环境与真实环境。这些创新使具身智能体具备全方位感知复杂环境的能力，能自然地与人类互动，并可靠地执行任务。

尽管学界对挖掘多层机器学习（MLMs）强大的感知与推理能力兴趣浓厚，但目前仍缺乏一项全面的调研来梳理现有的具身人工智能研究、面临的挑战及未来研究方向。在MLMs时代，我们旨在通过系统性地调研从网络空间到物理世界中的具身人工智能发展，填补这一空白。本研究从多个维度展开：涵盖具身机器人、模拟器、四项代表性具身任务（视觉主动感知、具身交互、多模态智能体及仿真-现实适应），以及未来研究方向。我们相信，这项调研将清晰展现当前研究成果，并为这一新兴且极具前景的研究领域指明进一步发展的方向。

与先前研究的差异：尽管已有若干关于具身人工智能的综述论文[5]、[6]、[17]、[18]，但其中大部分已过时，因它们发表于多模态语言模型（MLMs）时代之前（该时代

始于2023年左右）。据我们所知，2023年后仅有两篇综述论文[6]、[18]分别聚焦于视觉-语言-动作模型与面向智能制造的具身人工智能系统。配备AI大脑、身体及跨模态传感器的具身人工智能概念最早见于[18]，该研究亦是首个为基础模型时代未来智能制造提出具身人工智能系统技术架构的成果。然而，既往综述尚未充分考虑MLMs、WMs及具身智能体等要素，且对具身机器人与仿真器的最新发展也有所忽视。为解决这一快速发展领域中综合性综述论文匮乏的问题，我们提出了本项综合性综述，其内容涵盖了具有代表性的具身机器人、模拟器以及四项主要研究任务：具身感知、具身交互、具身智能体以及模拟到现实的适应性。总之，本研究的主要贡献有三个方

- 据我们所知，这是首项基于多层模型（MLMs）和权重矩阵（WMs）从网络空间与物理空间对齐角度对具身人工智能进行的综合性研究，为方法论、评估标准、挑战及应用领域提供了新颖见解。
- 我们将具身人工智能划分为机器人、模拟器等核心组成部分，并归纳为四大研究方向：具身感知、具身交互、具身智能体以及仿真到现实的适应性转换，这些构成了具身人工智能的详细分类体系。
- 为促进稳健通用具身智能体的研发，我们提出了一种新的数据集标准 ARIO（AllRobotsInOne）和统一的大规模 ARIO 数据集，该数据集包含约300万集数据，涵盖258个系列和321,064项任务。

本调查报告其余部分结构安排如下：第二章介绍具身机器人技术；第三章阐述通用型与真实场景具身模拟器；第四章解析具身感知机制，涵盖主动视觉感知与视觉语言导航技术；第五章探讨具身交互系统；第六章重点介绍具身智能体，包括多模态基础模型与任务规划算法；第七章阐述仿真到现实的适配策略，涉及具身世界模型构建、数据采集与训练方法；第八章则展望具有发展潜力的研究方向。

II.有形机器人

具身智能体与物理环境进行交互，包括机器人、智能家电和自动驾驶车辆等。如图2(a)所示，固定底座机器人因其精度高而被广泛应用于实验室自动化和工业领域，例如Franka Emika熊猫机器人[19]、[20]，Kuka iiwa机器人[21]、[22]，以及Sawyer机器人[23]、[24]。图2(b)展示的轮式机器人凭借其结构简单、成本低廉，在物流和仓储领域表现出高效性能，例如Kiva和Jackal机器人[25]；但它们在复杂地形环境中面临挑战。如图所示的履带式机器人.....

TABLE II

GENERAL SIMULATOR. **HFPS**: HIGH-FIDELITY PHYSICAL SIMULATION; **HQGR**: HIGH-QUALITY GRAPHICS RENDERING; **RRL**: RICH ROBOT LIBRARY; **DLS**: DEEP LEARNING SUPPORT; **LSPC**: LARGE-SCALE PARALLEL COMPUTING; **ROS**: TIGHT INTEGRATION WITH ROS; **MSS**: MULTIPLE SENSOR SIMULATION; **CP**: CROSS-PLATFORM **NAV**: ROBOT NAVIGATION **AD**: AUTO DRIVING; **RL**: REINFORCEMENT LEARNING **LSPS**: LARGE-SCALE PARALLEL SIM **MR**: MULTI-ROBOT SYSTEMS **RS**: ROBOT SIMULATION. ○ INDICATES THAT THE SIMULATOR EXCELS AT THIS ASPECT.

Simulator	Year	HFPS	HQGR	RRL	DLS	LSPC	ROS	MSS	CP	Physics Engine	Main Applications
Genesis [35]	2024	○	○	○	○	○	○	○	○	Custom	RL, LSPS, RS
Isaac Sim [36]	2023	○	○	○	○	○	○	○	○	PhysX	Nav, AD
Isaac Gym [37]	2019	○	○	○	○	○	○	○	○	PhysX	RL, LSPS
Gazebo [38]	2004	○	○	○	○	○	○	○	○	ODE, Bullet, Simbody, DART	Nav, MR
PyBullet [39]	2017	○	○	○	○	○	○	○	○	Bullet	RL, RS
Webots [40]	1996	○	○	○	○	○	○	○	○	ODE	RS
MuJoCo [41]	2012	○	○	○	○	○	○	○	○	Custom	RL, RS
Unity ML-Agents [42]	2017	○	○	○	○	○	○	○	○	Custom	RL, RS
AirSim [43]	2017	○	○	○	○	○	○	○	○	Custom	Drone sim, AD, RL
MORSE [44]	2015	○	○	○	○	○	○	○	○	Bullet	Nav, MR
V-REP (CoppeliaSim) [45]	2013	○	○	○	○	○	○	○	○	Bullet, ODE, Vortex, Newton	MR, RS

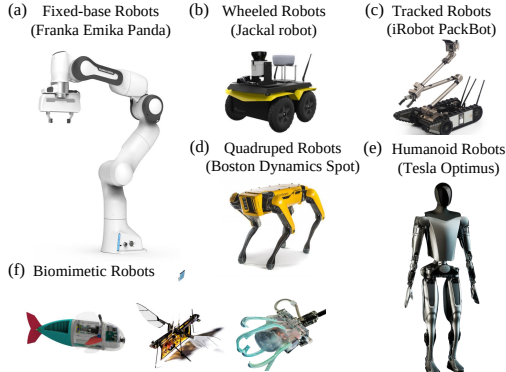


Fig. 2. The Embodied Robots include Fixed-base Robots, Quadruped Robots, Humanoid Robots, Wheeled Robots, Tracked Robots, and Biomimetic Robots.

Fig. 2 (c), are ideal for off-road tasks such as agriculture and disaster recovery. Their track systems provide stability on soft terrains [26]. Quadruped robots, illustrated in Fig. 2 (d), excel in complex terrain exploration and rescue missions. Examples include Unitree Robotics' A1 and Go1, and Boston Dynamics Spot. Humanoid robots, shown in Fig. 2 (e), mimic human movements and behaviors to provide personalized services. Their dexterous hands enable them to perform intricate tasks [27], [28]. With LLM, these robots are anticipated to enhance efficiency and safety in manufacturing, healthcare, and services [29]. Biomimetic robots, depicted in Fig. 2 (f), replicate the movements and functions of natural organisms. This simulation aids in operating within complex environments and boosts energy efficiency by emulating biological mechanisms [30], [31]. Examples include fish-like [32], insect-like [33], and soft-bodied robots [34].

III. EMBODIED SIMULATORS

Embodied simulators are crucial for embodied AI due to their cost-effectiveness, safety features, scalability, rapid prototyping, and accessibility for research. They allow for controlled experimentation, data generation for training and evaluation, and standardized benchmarks. To facilitate interaction with the environment, it's essential to build realistic simulations by considering physical characteristics, object properties, and their interactions. This section will introduce the commonly used simulation platforms in two parts: the general simulator based on underlying simulation and the simulator based on real scenes.

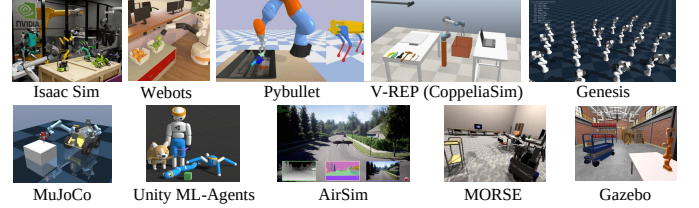


Fig. 3. Examples of General Simulators. The MuJoCo's figure is from [46].

A. General Simulator

The physical interactions and dynamic changes present in real environments are irreplaceable. However, deploying embodied models in the physical world often incurs high costs and faces numerous challenges. General-purpose simulators provide a virtual environment that closely mimics the physical world, allowing for algorithm development and model training, which offers significant cost, time, and safety advantages.

Isaac Sim [36] is an advanced simulation platform for robotics and AI research. It has high-fidelity physical simulation, real-time ray tracing, an extensive library of robotic models, and deep learning support. Its application scenarios include autonomous driving, industrial automation, and human-robot interaction. **Gazebo** [47] is an open-source simulator for robotics research. It has extensive robot libraries, and tight integration with ROS. It supports the simulation of various sensors and offers numerous pre-built robot models and environments. It is mainly used for robot navigation and control and multi-robot systems. **PyBullet** [39] is the python interface for the Bullet physics engine. It is easy to be used and has diverse sensor simulation and deep learning integration. PyBullet supports real-time physical simulation, including rigid body dynamics, collision detection, and constraint solving. Moreover, the newly launched **Genesis** [35] has differentiable physics engine and impressive generative capabilities. Table. II presents the key features and primary application scenarios of 11 general-purpose simulators. Fig. 3 shows the visualization effects of the general simulators.

B. Real-Scene Based Simulators

Achieving universal embodied agents in household activities is a primary focus in embodied AI. These embodied agents need to deeply understand human daily life and perform complex embodied tasks such as navigation and interaction in indoor environments. To meet the demands of these complex tasks, the simulated environments need to be close to the

表II通用模拟器。HFPS: 高保真物理模拟; HQGR: 高质量图形渲染; RRL: 丰富机器人库; DLS: 深度学习支持; LSPC: 大规模并行计算; ROS: 与ROS紧密集成; MSS: 多传感器
模拟; CP: 跨平台; NAV: 机器人导航; AD: 自动驾驶; RL: 强化学习; LSPS: 大规模并行仿真; Mr: 多机器人系统; RS: 机器人模拟。○表示该模拟器在该方面表现卓越。

模拟器	年份	HFPS	HQGR	RRL	DLS	LSPC	ROS	MSS	CP	物理引擎	主要应用
《创世纪》[35]	2024	○	○	○	○	○	○	○	○	自定义	RL, LSPS, RS
艾萨克·西姆[36]	2023	○	○	○	○	○	○	○	○	PhysX	Nav, AD
艾萨克体育馆[37]	2019	○	○	○	○	○	○	○	○	PhysX	RL, LSPS
凉亭[38]	2004	○	○	○	○	○	○	○	○	ODE、Bullet、Simbody、DART	Nav先生
PyBullet [39]	2017	○	○	○	○	○	○	○	○	子弹	RL, RS
Webots [40]	1996	○	○	○	○	○	○	○	○	子弹	RS
MuJoCo [41]	2012	○	○	○	○	○	○	○	○	自定义	RL, RS
Unity ML-Agent[42]	2017	○	○	○	○	○	○	○	○	自定义	RL, RS
AirSim [43]	2017	○	○	○	○	○	○	○	○	自定义	无人飞机模拟、自主决策、强化学习
摩尔斯[44]	2015	○	○	○	○	○	○	○	○	子弹	Nav先生
V-REP (CoppeliaSim) [45]	2013	○	○	○	○	○	○	○	○	子弹、常微分方程、涡旋、牛顿	MR, RS

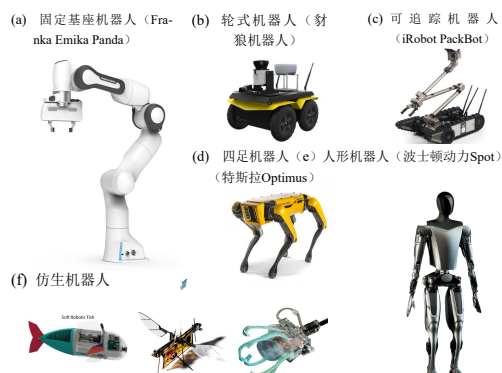


图2. 具身机器人包括固定底座机器人、四足机器人、人形机器人、轮式机器人、履带式机器人和仿生机器人。

图2(c)所示机器人非常适合农业和灾害恢复等越野任务。其履带系统能在软质地形上提供稳定性[26]; 图2(d)展示的四足机器人在复杂地形探索和救援任务中表现卓越, 例如Unitree Robotics的A1和Go1以及Boston Dynamics Spot。图2(e)所示的人形机器人通过模仿人类动作与行为提供个性化服务, 其灵巧的手部使其能够执行复杂任务[27]、[28]。借助大语言模型, 这些机器人有望提升制造业、医疗保健及服务领域的效率与安全性[29]; 图2(f)所示的仿生机器人则复制了天然生物的运动与功能, 这种模拟有助于在复杂环境中操作, 并通过模仿生物机制提升能效[30]、[31], 典型实例包括鱼形机器人[32]、昆虫形机器人[33]及软体机器人[34]。

III. 体化模拟器

具身模拟器对于具身人工智能至关重要, 这得益于其成本效益高、安全性强、可扩展性好、支持快速原型开发以及便于科研应用等特点。它们能够实现受控实验、为训练和评估生成数据, 并提供标准化的基准测试。为了与环境的有效交互, 构建逼真的模拟环境时必须充分考虑物理特性、物体属性及其相互作用。本节将分两部分介绍常用的模拟平台: 基于底层模拟技术的通用模拟器, 以及基于真实场景的模拟器。

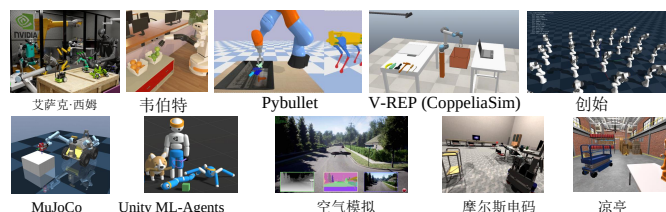


图3. 通用模拟器示例。MuJoCo的示意图引自[46]。

A. 通用模拟器

真实环境中存在的物理交互与动态变化具有不可替代性。然而, 在物理世界中部署具身模型往往面临高昂成本与诸多挑战。通用模拟器通过构建高度还原物理世界的虚拟环境, 既能支持算法开发与模型训练, 又能在成本控制、时间效率及安全性方面带来显著优势。

Isaac Sim[36]是一个用于机器人与人工智能研究的先进仿真平台。它具备高保真物理仿真、实时光线追踪、丰富的机器人模型库以及深度学习支持。其应用场景包括自动驾驶、工业自动化和人机交互。Gazebo[47]是一款开源机器人研究仿真器。它拥有丰富的机器人库, 并与ROS实现紧密集成。该平台支持多种传感器仿真, 并提供大量预构建机器人模型与环境。主要应用于机器人导航控制及多机器人系统。PyBullet[39]是Bullet物理引擎的Python接口。操作简便且具备多样化的传感器仿真与深度学习集成能力。PyBullet支持实时物理仿真, 包括刚体动力学、碰撞检测与约束求解。此外, 新推出的Genesis[35]拥有可微分物理引擎与卓越的生成能力。表。图II展示了11款通用模拟器的关键特性及主要应用场景。图3呈现了这些通用模拟器的可视化效果。

B. 基于真实场景的仿真器

实现家庭活动中普遍存在的具身智能体是具身人工智能的核心目标。这类智能体需深度理解人类日常生活, 并能在室内环境中完成导航、交互等复杂具身任务。为满足这些复杂任务的需求, 模拟环境必须高度贴近真实场景。

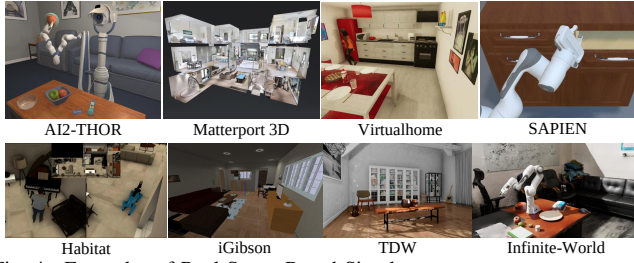


Fig. 4. Examples of Real-Scene Based Simulators.

real world, which places high demands on the complexity and realism of the simulators. These simulators mostly collect data from the real world, create photorealistic 3D assets, and build scenes using 3D game engines like UE5 and Unity. The rich and realistic scenes make simulators based on real world environments the top choice in household activities.

SAPIEN [48] stands out for its design, specifically tailored to simulate interactions with joint objects like doors, cabinets, and drawers. **VirtualHome** [49] is notable for its unique environment graph, which facilitates high-level embodied planning based on natural language descriptions of environments. While **AI2-ThOR** [50] offers a wealth of interactive scenes, these interactions, similar to those in VirtualHome, are script-based and lack real physical interactions. This design suffices for embodied tasks not requiring fine-grained interactions. Both **iGibson** [51] and **TDW** [52] provide fine-grained embodied control and highly simulated physical interactions. iGibson excels in offering abundant and realistic large-scale scenes, making it suitable for complex and long-term mobile operations, whereas TDW allows greater user freedom in scene expansion and features unique audio and flexible fluid simulations, making it indispensable for related simulation scenarios. **Matterport3D** [53], a foundational 2D-3D visual dataset, is widely used and extended in embodied AI benchmarks. Although the embodied agent in Habitat lacks interaction capabilities, its extensive indoor scenes, user-friendly interfaces, and open framework make it highly regarded in embodied navigation. **InfiniteWorld** [54] focuses on unified and scalable simulation framework and implemented various improvements and the latest implicit asset reconstruction, as well as natural language-driven scene generation and editing. It provides strong support for complex robotic interactions through distributed collaboration, AI assistance, and Human-in-the-Loop.

Besides, automated simulation scene construction is greatly beneficial for obtaining high-quality embodied data. **RoboGen** [55] customizes tasks from randomly sampled 3D assets through LLMs, thereby creating scenes and automatically training agents; **HOLODECK** [56] can automatically customize corresponding high-quality simulation scenes in AI2-THOR based on human instructions; **PhyScene** [57] generates interactive and physically consistent high-quality 3D scenes based on conditional diffusion. The Allen Institute for Artificial Intelligence expanded AI2-THOR and proposed **ProcTHOR** [58], which can automatically generate simulated scenes with sufficient interactivity, diversity, and rationality.

IV. EMBODIED PERCEPTION

The “north stars” of the future of visual perception is embodied-centric visual reasoning and social intelligence [59].

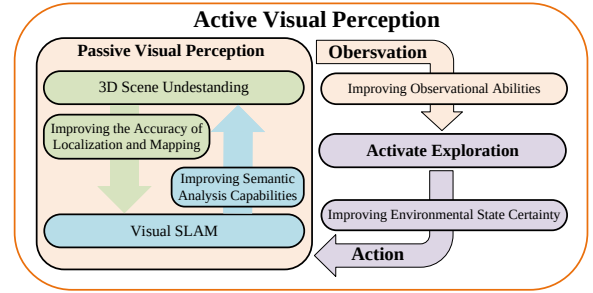


Fig. 5. The schematic diagram of active visual perception. Visual SLAM and 3D Scene Understanding provide the foundation for passive visual perception, while active exploration provides activeness to the passive perception system. These elements work collaboratively for the active visual perception system.

Unlike merely recognizing objects in images, agent with embodied perception must move in the physical world and interact with the environment. This requires a deeper understanding of 3D space and dynamic environments. Embodied perception requires visual perception and reasoning, understanding the 3D relations within a scene, and predicting and performing complex tasks based on visual information.

A. Active Visual Perception

Active visual perception systems require fundamental capabilities such as state estimation, scene perception, and environment exploration. As shown in Fig. 5, these capabilities have been extensively studied within the domains of Visual Simultaneous Localization and Mapping (vSLAM) [94], [95], 3D Scene Understanding [96], and Active Exploration [11]. These research areas contribute to developing robust active visual perception systems, facilitating improved environmental interaction and navigation in complex, dynamic settings. We briefly introduce these three components and summarize the methods mentioned in each part in Table III.

1) *Visual Simultaneous Localization and Mapping*: Simultaneous Localization and Mapping (SLAM) aims to determine a robot’s position within an unknown environment while simultaneously constructing a map of the environment [97]. Range-based SLAM [98], [99] relies on rangefinders, such as laser scanners, radar, and sonar, to generate point cloud representations. However, this approach is costly and provides limited environmental information. In contrast, Visual SLAM (vSLAM) [94], [95] employs on-board cameras to capture image frames and build environmental representations. Its advantages include low hardware costs, high accuracy in small-scale scenarios, and the ability to capture rich environmental details. Classical vSLAM can be broadly categorized into Traditional vSLAM and Semantic vSLAM [95]. Traditional vSLAM uses image data and multi-view geometry to estimate a robot’s pose and construct low-level maps (e.g., sparse, semi-dense, or dense point clouds) through methods like filter-based approaches (e.g., MonoSLAM [60]), keyframe-based methods (e.g., ORB-SLAM [61]), and direct tracking techniques (e.g., LSD-SLAM [62]). However, low-level maps do not directly correspond to objects, making them challenging for robots to interpret and utilize. Semantic vSLAM addresses this limitation by integrating semantic information, enhancing robots’ ability to perceive and navigate unexplored environments.

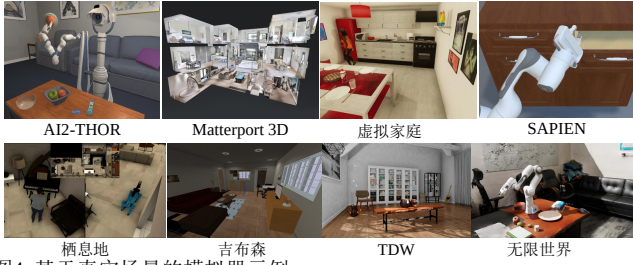


图4. 基于真实场景的模拟器示例。

现实世界对模拟器的复杂度与真实感提出了严苛要求。这类模拟器主要采集现实世界数据，生成逼真的3D虚拟资产，并运用UE5、Unity等3D游戏引擎构建场景。凭借丰富细腻的场景表现，基于真实环境打造的模拟器已成为家庭日常活动的首选方案。

SAPIEN[48]以其设计脱颖而出，专门用于模拟与门、橱柜和抽屉等联合对象的交互。**VirtualHome**[49]以其独特的环境图谱著称，能够基于环境的自然语言描述实现高层次的具身规划。虽然**AI2-ThOR**[50]提供了丰富的交互场景，但这些交互与**VirtualHome**类似，基于脚本且缺乏真实的物理交互。该设计足以满足不需要精细交互的具身任务需求。**iGibson**[51]与**TDW**[52]均提供精细的具身控制和高度模拟的物理交互。**iGibson**在提供丰富逼真的大规模场景方面表现卓越，适用于复杂且长期的移动操作场景；而**TDW**则允许用户更自由地扩展场景，并具备独特的音频效果和灵活的流体模拟特性，使其成为相关模拟场景中不可或缺的工具。**Matterport3D**[53]作为基础性的二维-三维视觉数据集，在具身人工智能基准测试中被广泛使用并不断扩展。尽管**Habitat**中的具身智能体缺乏交互能力，但其丰富的室内场景、用户友好的界面以及开放框架使其在具身导航领域备受推崇。**InfiniteWorld**[54]专注于统一且可扩展的仿真框架，实现了多项改进及最新的隐式资产重建技术，同时支持自然语言驱动的场景生成与编辑。该平台通过分布式协作、人工智能辅助及人机交互模式，为复杂的机器人交互提供了有力支持。

此外，自动化模拟场景构建对于获取高质量的具身数据具有显著优势。**RoboGen**[55]通过大型语言模型从随机采样的3D资产中定制任务，从而创建场景并自动训练智能体；**holodeck**[56]可根据人类指令在**AI2-THOR**平台上自动生成相应的高质量模拟场景；**PhyScene**[57]基于条件扩散技术生成交互性强且物理一致的高质量3D场景。艾伦人工智能研究院扩展了**AI2-THOR**，并提出了**ProCTHOR**[58]，该系统能自动生成具备充分交互性、多样性和合理性的模拟场景。

IV. 具身感知

视觉感知未来发展的“北极星”体现为具身中心视觉推理与社会智能[59]。

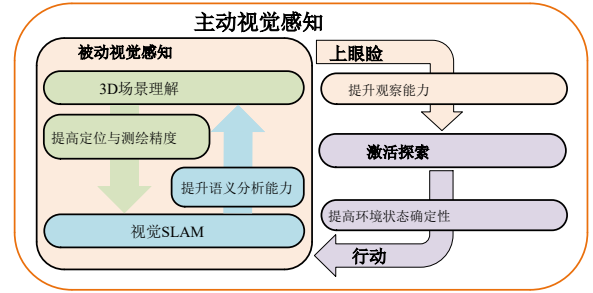


图5. 主动视觉感知示意图。视觉SLAM与三维场景理解为基础视觉感知奠定基石，主动探索则为被动感知系统注入活力。这些要素协同作用，共同构成主动视觉感知系统。

与仅在图像中识别物体不同，具有具身感知能力的智能体必须在物理世界中移动并与环境互动。这需要对三维空间及动态环境有更深入的理解。具身感知要求视觉感知与推理能力，包括理解场景中的三维关系，并基于视觉信息预测和执行复杂任务。

A. 主动视觉感知

主动视觉感知系统需要具备状态估计、场景感知和环境探索等基础能力。如图5所示，这些能力已在视觉同步定位与地图构建（vSLAM）[94]、[95]、三维场景理解[96]以及主动探索[11]等领域得到广泛研究。这些研究方向有助于开发稳健的主动视觉感知系统，从而在复杂动态环境中提升环境交互与导航能力。本文将简要介绍这三个核心组件，并在表III中对各部分提及的方法进行总结。

1) **视觉同步定位与地图构建**：同步定位与地图构建（SLAM）旨在确定机器人在未知环境中的位置，同时构建环境地图[97]。基于距离的SLAM[98]、[99]依赖激光扫描仪、雷达和声纳等测距设备生成点云表征。然而该方法成本高昂且环境信息获取有限。相比之下，视觉SLAM（vSLAM）[94]、[95]采用车载摄像头捕捉图像帧并构建环境表征。其优势包括硬件成本低、小规模场景中高精度以及能够捕捉丰富的环境细节。经典vSLAM可大致分为传统vSLAM与语义vSLAM[95]。传统vSLAM利用图像数据和多视角几何信息通过滤波方法（如MonoSLAM[60]）、关键帧方法（如ORB-SLAM[61]）以及直接跟踪技术（如LSD-SLAM[62]）。然而，低层级地图无法直接对应物体，这使得机器人难以解读和利用这些地图。语义视觉SLAM通过整合语义信息解决了这一局限性，显著提升了机器人在未探索环境中的感知与导航能力。

TABLE III
THE COMPARISON OF THE ACTIVE VISUAL PERCEPTION METHODS.

Function	Type	Methods
vSLAM	Traditional vSLAM	MonoSLAM [60], ORB-SLAM [61], LSD-SLAM [62]
	Semantic vSLAM	SLAM++ [63], QuadricSLAM [64], So-SLAM [65], SG-SLAM [66], OVD-SLAM [67], GS-SLAM [68]
3D Scene Understanding	Projection-based	MV3D [69], PointPillars [70], MVCNN [71]
	Voxel-based	VoxNet [72], SSCNet [73], MinkowskiNet [74], SSCNs [75], Embodiedscan [76]
	Point-based	PointNet [77], PointNet++ [78], PointMLP [79], PointTransformer [80], Swin3d [81], PT2 [82], 3D-VisTA [83], LEO [84], PQ3D [85], PointMamba [86], Mamba3D [87]
Active Exploration	Interacting with the environment	Pinto et al. [88], Tatiya et al. [89]
	Changing the viewing direction	Jayaraman et al. [90], NeU-NBV [91], Hu et al. [92], Fan et al. [93]

2) *3D Scene Understanding*: 3D scene understanding [100] aims to distinguish objects' semantics, identify their locations, and infer the geometric attributes from 3D scene data [101], which is fundamental in autonomous driving [102], robot navigation [103], and human-computer interaction [104] etc. A scene may be recorded as 3D point clouds using 3D scanning tools like LiDAR or RGB-D sensors. Unlike images, point clouds are sparse, disordered, and irregular. Recent advances in deep learning for 3D scene understanding can be categorized into projection-based, voxel-based, and point-based methods. Concretely, projection-based methods (e.g., MV3D [69], PointPillars [70], MVCNN [71]) project 3D points onto various image planes and employ 2D CNN-based backbones for feature extraction. Voxel-based methods convert point clouds into regular voxel grids to facilitate 3D convolution operations (e.g., VoxNet [72], SSCNet [73]), and some works improve their efficiency through sparse convolution (e.g., MinkowskiNet [74], SSCNs [75], Embodiedscan [76]). In contrast, point-based methods process point clouds directly (e.g., PointNet [77], PointNet++ [78], PointMLP [79]). Recently, to achieve model scalability, Transformers-based (e.g., PointTransformer [80], Swin3d [81], PT2 [82], 3D-VisTA [83], LEO [84], PQ3D [85]) and Mamba-based (e.g., PointMamba [86], Mamba3D [87]) architectures have emerged. Notably, PQ3D [85] enhances scene understanding by seamlessly integrating features from point clouds, multi-view images, and voxels.

3) *Active Exploration*: The 3D scene understanding methods allow robots to passively perceive the environment, with static information acquisition and decision-making regardless of scene changes. Thus, while passive perception is essential, it must be complemented by active exploration, enabling robots to dynamically interact with and perceive their surroundings. The relationship between them is shown in Fig. 5. Current methods addressing active perception focus on interacting with the environment [88], [89] or by changing the viewing direction to obtain more visual information [90]–[93].

For example, Pinto et al. [88] proposed a curious robot that learns visual representations through physical interaction with the environment rather than relying solely on dataset category labels. To address the challenge of interactive object perception across robots with varying morphologies, Tatiya et al. [89] proposed a multi-stage projection framework that transfers implicit knowledge through learned exploratory interactions. Recognizing the challenge of autonomously capturing informative observations, Jayaraman et al. [90] proposed a reinforcement learning method where an agent learns to actively acquire informative visual observations by reducing its uncertainty about unobserved parts of its environment. NeU-NBV

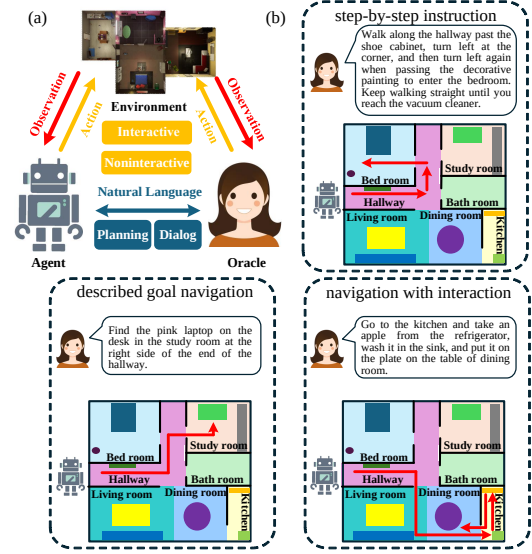


Fig. 6. (a) Overview of VLN. The embodied agent communicates with humans through natural language. Humans issue instructions to the embodied agent, who completes tasks such as planning and dialog. Subsequently, through collaborative cooperation or the embodied agent's independent actions, actions are made in interactive or non-interactive environments based on visual observations and instructions, (b) Different tasks of VLN.

[91] introduced a mapless planning framework that iteratively positions an RGB camera to capture the most informative images of an unknown scene. Hu et al. [92] developed a robot exploration algorithm that predicts the value of future states using a state value function. To address the issue of accidental input in open-world environments, Fan et al. [93] treated active recognition as a sequential evidence-gathering process, providing step-by-step uncertainty quantification and reliable prediction under evidence combination theory.

B. Visual Language Navigation

Visual Language Navigation (VLN) is an essential task, aiming at navigating in unseen environments following linguistic instructions. VLN requires robots to understand complex and diverse visual observations and meanwhile interpret instructions at different granularities. The input typically consists of two parts: visual information and natural language instructions. The visual information can either be a video of past trajectories or a set of historical-current observation images. The natural language instructions include the target that the agent needs to reach or the task that the agent is expected to complete. The agent must use the above information to select one or a series of actions from a list of candidates to fulfill the requirements of the natural language instructions. This process could be represented as $Action = \mathcal{M}(O, H, I)$, where $Action$ is the chosen action or a list of action candidates, O is the

表 III
主动视觉感知方法的比较

函数	类型	方法
vSLAM	传统视觉SLAM	MonoSLAM[60]、ORB-SLAM[61]、LSD-SLAM[62]
	语义vSLAM	SLAM++[63]、QuadricSLAM[64]、So-SLAM[65] SG-SLAM[66]、OVD-SLAM[67]、GS-SLAM[68]
3D场景理解	基于投影的	MV3D [69]、PointPillars [70]、MVCNN [71]
	基于体素的	VoxNet [72]、SSCNet [73]、MinkowskiNet [74]、SSCNs [75]、Embodiedscan [76]
	基于点的	PointNet[77]、PointNet++[78]、PointMLP[79]、PointTransformer[80]、Swin3d[81]、PT2[82]、3D-VisTA[83]、LEO[84]、PQ3D[85]、PointMamba[86]、Mamba3D[87]
主动勘探	与环境相互作用	Pinto等人[88]、Tatiya等人[89]
	改变观察方向	Jayaraman等人[90]、NeU-NBV[91]、Hu等人[92]、Fan等人[93]

2) **三维场景理解**：三维场景理解[100]旨在区分物体的语义、识别其位置，并从三维场景数据中推断几何属性[101]，这在自动驾驶[102]、机器人导航[103]及人机交互[104]等领域具有基础性意义。场景可通过激光雷达或RGB-D传感器等三维扫描工具以三维点云形式记录。与图像不同，点云具有稀疏性、无序性和不规则性特征。近期三维场景理解领域的深度学习进展可分为基于投影、基于体素及基于点的方法。具体而言，基于投影的方法（如MV3D[69]、PointPillars[70]、MVCNN[71]）将三维点投影至多个图像平面，并采用基于二维CNN的骨干网络进行特征提取；基于体素的方法将点云转换为规则体素网格以支持三维卷积运算（如VoxNet[72]、SSCNet[73]），部分研究通过稀疏卷积技术提升其效率（例如，MinkowskiNet[74]、SSCNs[75]、Embodiedscan[76]）。相比之下，基于点的方法直接处理点云（例如PointNet[77]、PointNet++[78]、PointMLP[79]）。近年来，为实现模型可扩展性，基于Transformer（例如PointTransformer[80]、Swin3d[81]、PT2[82]、3D-VisTA[83]、LEO[84]、PQ3D[85]）和基于Mamba（例如PointMamba[86]、Mamba3D[87]）的架构应运而生。值得注意的是，PQ3D[85]通过无缝整合点云、多视图图像和体素的特征来增强场景理解能力。

3) **主动探索**：三维场景理解方法使机器人能够被动感知环境，通过静态信息采集和决策机制，不受场景变化的影响。因此，虽然被动感知至关重要，但必须辅以主动探索，使机器人能够动态地与周围环境互动并感知环境。二者之间的关系如图5所示。当前针对主动感知的方法主要侧重于与环境交互[88]、[89]，或通过改变观察方向以获取更多视觉信息[90]–[93]。

例如，Pinto等人[88]提出了一种创新型机器人，其通过与环境的物理交互而非单纯依赖数据集类别标签来学习视觉表征。为应对形态各异机器人间交互式物体感知的挑战，Tatiya等人[89]提出了一种多阶段投影框架，通过学习性探索性交互传递隐性知识。针对自主捕获信息性观测数据的难题，Jayaraman等人[90]提出强化学习方法，使智能体通过降低对环境未观测区域的不确定性来主动获取信息性视觉观测。NeU-NBV[91]引入无地图规划框架，通

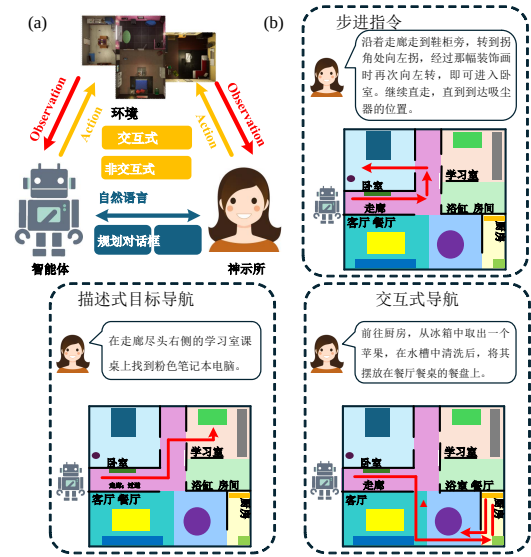


图6. (a) VLN 的概述。具身智能体通过自然语言与人类进行交互。人类向具身智能体发出指令，后者负责执行规划、对话等任务；随后，通过协作或独立行动，智能体在交互式或非交互式环境中根据视觉观察和指令执行相应操作。(b) VLN 的不同任务。

过迭代调整RGB相机位置以捕捉未知场景中最具信息量的图像。Hu等人[92]开发了基于状态值函数预测未来状态价值的机器人探索算法。为解决开放世界环境中的意外输入问题，Fan等人[93]将主动识别视为一个顺序性的证据收集过程，在证据组合理论框架下实现了逐步的不确定性量化与可靠的预测。

B. 视觉语言导航

视觉语言导航（VLN）是一项关键任务，旨在根据语言指令在未见过的环境中进行导航。VLN 要求机器人能够理解复杂多样的视觉观测信息，并同时以不同粒度解读指令。输入通常包含两部分：视觉信息和自然语言指令。视觉信息可以是过去的轨迹视频，或一组历史-当前的观测图像；自然语言指令则包括智能体需要到达的目标或预期完成的任务。智能体必须利用上述信息，从候选动作列表选择一个或一系列动作以满足自然语言指令的要求。该过程可表示为 $动作 = M(O, H, I)$ ，其中 $动作$ 是选定的动作或动作候选列表， O 是

TABLE IV

COMPARISON OF DIFFERENT VLN DATASETS. **M3D**: MATTERPORT3D, **AT**: AI2-THOR, **OG**: OMNIGIBSON, **I**: INDOOR, **D**: DISCRETE, **O**: OUTDOOR, **C**: CONTINUOUS, **SbS**: STEP-BY-STEP INSTRUCTIONS, **DGN**: DESCRIBED GOAL NAVIGATION, **DDN**: DEMAND-DRIVEN NAVIGATION, **NwI**: NAVIGATION WITH INTERACTION, **LSNwI**: LONG-SPAN NAVIGATION WITH INTERACTION, **D&O**: DIALOG AND ORACLE

Dataset	Year	Simulator	Environment	Feature	Size
R2R [105]	2018	M3D	I, D	SbS	21,567
R4R [106]	2019	M3D	I, D	SbS	200,000+
VLN-CE [107]	2020	Habitat	I, C	SbS	-
TOUCHDOWN [108]	2019	-	O, D	SbS	9,326
REVERIE [109]	2020	M3D	I, D	DGN	21,702
SOON [110]	2021	M3D	I, D	DGN	3,848
DDN [111]	2023	AT	I, C	DDN	30,000+
ALFRED [112]	2020	AT	I, C	NwI	25,743
OVMM [113]	2023	Habitat	I, C	NwI	7,892
BEHAVIOR-1K [114]	2023	OG	I, C	LSNwI	1,000
CVDN [115]	2020	M3D	I, D	D&O	2,050
DialFRED [116]	2022	AT	I, C	D&O	53,000

current observation, H is the historical information, and I is the natural language instruction.

1) *Datasets*: In VLN, natural language instructions can be a series of detailed action descriptions, a fully described goal, or just a roughly described task, even only the demands of human. The tasks that embodied agents need to complete maybe just a single navigation, or navigation with interaction, or multiple navigation tasks that need to be completed in sequence. These differences bring different challenges to VLN, and many different datasets have been built. Based on these differences, we introduce some important VLN datasets.

Room to Room (R2R) [105] is a VLN dataset based on Matterport3D. In R2R, embodied agents navigate according to step-by-step instructions, choosing the next adjacent navigation graph node to advance based on visual observations until they reach the target location. **Room-for-Room** [106] extends the paths in R2R to longer trajectories, which requires stronger long-distance instruction and history alignment capabilities of embodied agents. **VLN-CE** [107] extends R2R and R4R to continuous environments, embodied agents can move freely in the scene. Different from the above datasets based on indoor scenes, the **TOUCHDOWN** dataset [108] is created based on Google Street View. In TOUCHDOWN, embodied agents follow instructions to navigate in the street view rendering simulation of New York City to find the specified object. Similar to R2R, the **REVERIE** dataset [109] is also built based on the Matterport3D simulator. REVERIE requires embodied agents to accurately locate the distant invisible target object specified by concise, human-annotated high-level natural language instructions. In **SOON** [110], agents receive a long and complex instruction from coarse to fine to find the target object in the 3D environment. During navigation, agents first search a larger area, and then gradually narrow the search range according to the visual scene and instructions. **DDN** [111] moves a step further beyond these datasets, only providing human demands without specifying explicit objects. The agent needs to navigate through the scene to find objects.

ALFRED dataset [112] is based on the AI2-THOR simulator. In ALFRED, embodied agents need to understand environmental observations and complete household tasks in an interactive environment according to coarse-grained and fine-grained instructions. The task in **OVMM** [113] dataset is to pick any object in any unseen environment and place it

TABLE V
COMPARISON OF VLN METHODS.

Method	Model	Year	Feature
Memory-Understanding Based	LVERG [117]	2020	Graph Learning
	CMG [118]	2020	Adversarial Learning
	RCM [119]	2021	Reinforcement learning
	FILM [120]	2022	Semantic Map
	LM-Nav [121]	2022	Graph Learning
	HOP [122]	2022	History Modeling
	NaviLLM [123]	2024	Large Model
	FSTT [124]	2024	Test-Time Augmentation
	DiscussNav [125]	2024	Large Model
	GOAT [126]	2024	Causal Learning
Future-Prediction Based	VER [127]	2024	Environment Encoder
	NaVid [128]	2024	Large Model
	LookBY [129]	2018	Reinforcement Learning
	NvEM [130]	2021	Environment Encoder
	BGBL [131]	2022	Graph Learning
	Mic [132]	2023	Large Model
Others	HNR [133]	2024	Environment Encoder
	ETPNv [134]	2024	Graph Learning
	MCR-Agent [135]	2023	Multi-Level Model
	OVLM [136]	2023	Large Model

in a specified location. **OVMM** provides a simulation based on Habitat and a framework for implementation in the real world. **Behavior-1K** dataset [114] is based on human needs, comprising 1,000 long-sequence, complex, skill-dependent daily tasks. Agents need to complete long-span navigation-interaction tasks which contain thousands of low-level action steps based on visual information and language instructions. These complex tasks requires strong capabilities of understanding and memory. **CVDN** [115] requires embodied agents to navigate to the target based on dialogue history, and ask questions for help to decide the next action when uncertain. **DialFRED** [116], an extension of ALFRED, allows agents to ask questions during the navigation and interaction process to get help. These datasets introduce additional oracles, and embodied agents need to obtain more information beneficial to navigation by asking questions.

2) *Method*: VLN has made great strides recently with the astonishing performance of LLMs, the direction and focus of VLN have been profoundly influenced. Nevertheless, the VLN methods can be divided into two directions: **Memory-Understanding Based** and **Future-Prediction Based**.

Memory-Understanding based methods focus on the perception and understanding of the environment, as well as model design based on historical observations or trajectories, which is a method based on past learning. Future-Prediction based methods pay more attention to modeling, predicting, and understanding the future state, which is a method for future learning. Since VLN can be regarded as a partially observable Markov decision process, where future observations depend on the current environment and actions of the intelligent agent, historical information has important significance for navigation decisions, especially long-span navigation decisions, hence Memory-Understanding based methods have always been the mainstream of VLN. However, Future-Prediction based methods still have important significance. Its essential understanding of the environment has great value in VLN in continuous environments, especially with the rise of the concept of world model, Future-Prediction based methods are receiving more and more attention from researchers.

Memory-Understanding based. Graph-based learning is an essential part of the memory-understanding based method.

表 IV

不同 VLN 数据集的比较。M3D: MATTERPORT3D, ; AT: AI2-THOR; OG: OMNIGIBSON; I: 室内; D: 离散; O: OUTDOOR; 连续式; SBS: 逐步指令; DGN: 描述性目标导航; DDN: 需求驱动式导航; NWI: 交互式导航; LSNWI: 长跨度导航
导航与交互, **D&O**: 对话与Oracle

数据集	年	模拟器	环境	特征	尺寸
R2R [105]	2018	M3D	I, D	锑硫化物	21,567
R4R [106]	2019	M3D	I, D	锑硫化物	200,000+
VLN-CE [107]	2020	栖息地	I, C	锑硫化物	-
达阵[108]	2019	-	O, D	锑硫化物	9,326
遐想[109]	2020	M3D	I, D	DGN	21,702
很快[110]	2021	M3D	I, D	DGN	3,848
DDN [111]	2023	AT	I, C	数字数据服务	30,000+
阿尔弗雷德[112]	2020	AT	I, C	NwI	25,743
OVMM [113]	2023	栖息地	I, C	NwI	7,892
行为-1K[114]	2023	OG	I, C	LSNWI	1,000
CVDN [115]	2020	M3D	I, D	D&O	2,050
DialFRED [116]	2022	AT	I, C	D&O	53,000

当前观测值, H 为历史信息, I 为自然语言指令。

1) **数据集**: 在 VLN 中, 自然语言指令可以是详细的动作描述序列、完整的目标描述, 或是仅包含人类需求的粗略任务描述。智能体需要完成的任务可能仅涉及单次导航、带交互的导航, 或是需按顺序完成的多个导航任务。这些差异给 VLN 带来了不同挑战, 目前已构建了多种数据集。基于这些差异, 我们重点介绍若干重要 VLN 数据集。

Room to Room (R2R)[105]是一个基于Matterport3D的 VLN 数据集。在R2R中, 具身智能体根据逐步指令进行导航, 通过视觉观察选择下一个相邻导航图节点推进, 直至到达目标位置。**Room-for-Room**[106]将R2R中的路径扩展为更长的轨迹, 这需要具身智能体具备更强的远距离指令处理能力和历史信息对齐能力。**VLN -CE**[107]将R2R和R4R扩展至连续环境, 具身智能体可在场景中自由移动。与基于室内场景的上述数据集不同, **touchdown**数据集[108]基于Google街景创建。在touchdown任务中, 具身智能体遵循指令在纽约市街景渲染模拟环境中导航以定位指定物体。与R2R类似, **reverie**数据集[109]也基于Matterport3D模拟器构建。漫游智能需要具身智能体能够根据简洁的人类标注高级自然语言指令, 精准定位远处的不可见目标物体。在**SOON**[110]研究中, 智能体会从粗略到精细接收长而复杂的指令, 以在三维环境中定位目标物体。导航过程中, 智能体首先搜索较大区域, 随后根据视觉场景和指令逐步缩小搜索范围。**DDN**[111]研究则突破了传统数据集限制, 仅提供人类需求而不指定具体物体。智能体需要通过场景导航来发现目标物体。

alfred数据集[112]基于 AI2-THOR 模拟器。在 alfred 中, 具身智能体需要理解环境观察结果, 并根据粗粒度和细粒度指令在交互环境中完成家庭任务。**OVMM**[113]数据集的任务是在任何未见过的环境中选取任意物体并将其放置在指定位置。OVMM 提供了基于 Habitat 的模拟环境

表 V

VLN 方法比较

方法	模型	年	特征
记忆力理解基础	LVERG [117]	2020	图学习、对抗学习、强
	CMG [118]	2020	化学习、语义图谱、图
	RCM [119]	2021	学习、历史建模、大型
	电影[120]	2022	模型、测试时数据增
	LM-Nav [121]	2022	强、大型模型、因果学
	HOP [122]	2022	习、环境编码器、大型
	NaviLLM [123]	2024	模型
	FSTT [124]	2024	
	DiscussNav [125]	2024	
	GOAT [126]	2024	
未来-预测基础	VER [127]	2024	
	NaVid [128]	2024	
	LookBY [129]	2018	强化学习
	NvEM [130]	2021	环境编码器
	BGBL [131]	2022	图学习
	Mic [132]	2023	大型模型
其他	HNR [133]	2024	环境编码器
	ETPNav [134]	2024	图学习
	MCR 代理[135]	2023	多层模型
	OVLM [136]	2023	大型模型

以及在现实世界中实现的框架。**Behavior-1K**数据集[114]基于人类需求, 包含 1,000 项长序列、复杂且依赖技能的日常任务。智能体需要完成包含数千个基于视觉信息和语言指令的低级动作步骤的长跨度导航交互任务。这些复杂任务要求具备强大的理解与记忆能力。**CVDN**[115]要求具身智能体根据对话历史导航至目标位置, 并在不确定时通过提问获取帮助以决定下一步行动。**DialFRED**[116]是 alfred 的扩展版本, 允许智能体在导航和交互过程中提出问题以获取帮助。这些数据集引入了额外的预言机, 具身智能体需要通过提问来获取更多有助于导航的信息。

2) **方法**: 随着LLMs惊人的性能表现, VLN 近年来取得了巨大进展, 其发展方向和重点受到了深刻影响。然而, VLN 方法可分为两个方向: **基于记忆理解**和**基于未来预测**。

基于记忆理解的方法侧重于环境感知与理解能力, 以及基于历史观测或轨迹的模型设计, 这类方法属于基于过往学习的范畴。基于未来预测的方法则更注重对模型构建、状态预测及未来状态理解, 属于未来学习方法。由于智能 VLN 可被视为部分可观测的马尔可夫决策过程(其中未来观测结果取决于当前环境及智能体行为), 历史信息对导航决策(尤其是长距离导航决策)具有重要价值, 因此基于记忆理解的方法始终是 VLN 领域的主流方向。然而, 基于未来预测的方法仍具有重要价值。其对环境本质的理解在连续环境 VLN 中具有显著应用价值, 特别是在世界模型概念兴起的背景下, 这类方法正受到越来越多研究者的关注。

基于记忆理解。图论学习是基于记忆理解方法的重要组成部分。

It usually represents the navigation process in the form of a graph, where the information obtained by the agent at each time step is encoded as nodes of the graph. The agent obtains global or partial navigation graph information as a representation of the historical trajectory. LVERG [117] encoded the language information and visual information of each node separately, design a new language and visual entity relationship graph to model the inter-modal relationship between text and vision, and the intra-modal relationship between visual entities. LM-Nav [121] used a goal-conditioned distance function to infer connections between original observation sets and construct a navigation graph, and extracted landmarks from the instructions through a LLM. Although HOP [122] is not based on graph learning, it requires to model time-ordered information at different granularities, thereby achieving a deep understanding of historical trajectories and memories.

The navigation graph discretizes the environment, but concurrently understanding and encoding the environment is also important. FILM [120] used RGB-D observations and semantic segmentation to gradually build a semantic map from 3D voxels during the navigation. VER [127] quantified the physical world into structured 3D units through 2D-3D sampling, providing fine-grained geometric details and semantics.

Different learning schemes explore how to utilize historical trajectories and memories better. Through adversarial learning, CMG [118] alternated between imitation learning and exploration encouragement schemes, effectively strengthening the understanding of instructions and historical trajectories, shortening the difference between training and inference. GOAT [126] directly trained unbiased models through Backdoor Adjustment Causal Learning (BACL) and Frontdoor Adjustment Causal Learning (FACL), conducted contrastive learning with vision, navigation history, and their combination to instructions, enabling the agent to make fuller use of information. The enhanced cross-modal matching method proposed by RCM [119] used goal-oriented external rewards and instruction-oriented internal rewards to perform cross-modal grounding globally and locally and learns from its own historical good decisions through self-supervised imitation learning. FSTT [124] introduced TTA into VLN and optimizes the model in terms of gradients and model parameters at two scales of time steps and tasks, effectively improving model performance.

The specific application of large models in Memory-Understanding based methods is to understand the representation of historical memory and to understand the environment and tasks based on its extensive world knowledge. NaviLLM [123] integrated the historical observation sequence into the embedding space through the visual encoder, inputs the multi-modal information of the fusion encoding into the LLM and fine-tunes it, reaching the state-of-the-art on multiple benchmarks. NaVid [128] made improvements in the encoding of historical information, achieves different degrees of information retention on historical observations and current observations through different degrees of pooling. LH-VLN [137] proposed the NavGen platform, the long-horizon navigation benchmark, and the Multi-Granularity Dynamic Memory (MGDM) module to enhance task evaluation and model adaptability in dynamic environments.

Future-Prediction Based. Graph-based learning is also widely used in Future-Prediction based methods. BGBL [131] and ETPNav [134] used a similar method to design a waypoint predictor that can predict movable path points in a continuous environment based on the observation of the current navigation graph node. They aim to migrate complex navigation in a continuous environment to node-to-node navigation in a discrete environment, thereby bridging the performance gap from discrete environments to continuous environments.

Improving the understanding and perception of the future environment through environmental encoding is also one of the research directions for predicting and exploring the future. NvEM [130] used a theme module and a reference module to perform fusion encoding of neighbor views from the global and local perspectives. This is actually an understanding and learning of future observations. HNR [133] used a large-scale pre-trained hierarchical neural radiation representation model to directly predict the visual representation of the future environment rather than pixel-level images using three-dimensional feature space encoding, and builds a navigable future path tree based on the representation of the future environment. They predict the future environment from different levels, providing effective references for navigation decisions.

Some reinforcement learning methods are also applied to predict and explore future states. LookBY [129] employed reinforcement prediction to enable the prediction module to imitate the world and forecast future states and rewards. This allows the agent to directly map “current observations” and “predictions of future observations” to actions, achieving state-of-the-art performance at the time. The rich world knowledge and zero-shot performance of large models provide many possibilities for Future-Prediction based methods. MiC [132] required the LLM to directly predict the target and its possible location from the instructions and provides navigation instructions through the description of scene perception. This method requires LLMs to fully exert its ‘imagination’ and build an imagined scene through prompts.

In addition, there are some methods that both learn from the past and for the future. MCR-Agent [135] designed a three-layer action strategy, which requires the model to predict the target from the instructions, predict the pixel-level mask for the target to be interact, and learn from the previous navigation decision; OVLM [136] required the LLMs to predict the corresponding operations and landmark sequences for the instructions. During the navigation process, the visual language map will be continuously updated and maintained, and the operations will be linked to the waypoints on the map.

V. EMBODIED INTERACTION

Embodied interaction refer to scenarios where agents interact with humans and the environment in physical or simulated space. The typical embodied interaction tasks are Embodied Question Answering (EQA) and embodied grasping.

A. Embodied Question Answering

For EQA task, the agent needs to explore the environment from a first-person perspective to gather information necessary

该方法通常以图结构形式呈现导航过程,其中智能体在每个时间步获取的信息被编码为图中的节点。智能体通过获取全局或局部导航图信息来表征历史轨迹。LVERG[117]分别对每个节点的语言信息和视觉信息进行编码,设计了新的语言与视觉实体关系图来建模文本与视觉之间的跨模态关系,以及视觉实体之间的模态内关系。LM-Nav[121]采用目标条件距离函数推断原始观测集间的关联性并构建导航图,同时通过大语言模型(LLM)从指令中提取地标信息。尽管HOP[122]不基于图学习方法,但需要在不同粒度下对时间序列信息进行建模,从而实现对历史轨迹与记忆的深度理解。

导航图对环境进行离散化处理,但同步理解和编码环境信息同样至关重要。FILM[120]通过RGB-D观测数据与语义分割技术,在导航过程中逐步从三维体素构建语义地图。VER[127]则通过二维到三维的采样方法,将物理世界量化为结构化三维单元,从而提供精细的几何细节与语义信息。

不同的学习方案探索了如何更有效地利用历史轨迹和记忆。通过对抗学习,CMG[118]在模仿学习与探索激励方案之间交替切换,有效增强了对指令和历史轨迹的理解,缩短了训练与推理之间的差距。GOAT[126]采用后门调整因果学习(BACL)和前门调整因果学习(FACL)直接训练无偏模型,并结合视觉、导航历史及其组合与指令进行对比学习,使智能体能更充分地利用信息。RCM[119]提出的增强型跨模态匹配方法通过目标导向的外部奖励和指令导向的内部奖励,在全局和局部层面实现跨模态定位,并通过自监督模仿学习从自身历史正确决策中学习。FSTT[124]将TTA引入VLN模型,并从时间步长和任务两个维度优化模型的梯度和参数,显著提升了模型性能。

大型模型在基于记忆理解的方法中的具体应用,旨在通过其丰富的世界知识来理解历史记忆的代表方式,并基于此理解环境与任务。NaviLLM[123]通过视觉编码器将历史观测序列嵌入到嵌入空间,将融合编码的多模态信息输入大语言模型并进行微调,从而在多项基准测试中达到业界领先水平。NaVid[128]通过不同层级的池化技术改进了历史信息编码方式,实现了对历史观测与当前观测信息不同程度的保留。LHVLN[137]提出了NavGen平台、长时程导航基准测试以及多粒度动态记忆(MGDM)模块,以提升动态环境下的任务评估能力和模型适应性。

基于未来预测。图论学习方法在基于未来预测的方法中也得到广泛应用。BGBL[131]与ETPNav[134]采用类似方法设计了航路点预测器,该模型能够通过观测当前导航图节点,在连续环境中预测可移动路径点。其核心目标是将复杂连续环境导航任务迁移至离散环境中的节点间导航场景,从而弥合离散环境与连续环境之间的性能差距。

通过环境编码提升对未来环境的理解与感知能力,也是预测与探索未来发展的研究方向之一。NvEM[130]采用主题模块与参考模块,从全局与局部视角对邻近视图进行融合编码,这本质上是对未来观测数据的理解与学习过程。HNR[133]则运用大规模预训练分层神经辐射表征模型,通过三维特征空间编码直接预测未来环境的视觉表征(而非像素级图像),并基于未来环境表征构建可导航的未来路径树。该方法从多层级维度预测未来环境,为导航决策提供了有效参考依据。

一些强化学习方法也被应用于预测和探索未来状态。LookBY[129]采用了强化预测技术,使预测模块能够模拟现实世界并预测未来状态与奖励。这使得智能体能够直接将“当前观测结果”与“对未来观测结果的预测”映射为具体动作,在当时实现了最先进的性能表现。大型模型丰富的世界知识及其零样本学习能力,为基于未来预测的方法提供了广阔的应用前景。MiC[132]要求大语言模型直接根据指令预测目标及其可能位置,并通过场景感知描述生成导航指令。该方法要求大语言模型充分发挥其“想象力”,根据输入提示构建出想象中的场景。

此外,还存在一些既能从历史经验中学习又能面向未来发展的方法。MCR-Agent[135]设计了三层动作策略框架,该模型需要根据指令预测目标位置、预测目标交互所需的像素级掩码,并从前次导航决策中学习经验;OVLM[136]则要求大语言模型(LLM)预测指令对应的运算步骤和地标序列。在导航过程中,视觉语言地图将持续更新维护,各运算步骤将与地图上的航点进行关联。

V. 具身化互动

具身交互指智能体在物理或模拟空间中与人类及环境进行交互的场景。典型具身交互任务包括具身问答(EQA)和具身抓取。

A. 具身化问答研究

在EQA任务中,智能体需要以第一人称视角探索环境,以收集必要的信息。

TABLE VI
COMPARISON OF DIFFERENT EQA DATASETS.

Dataset	Year	Type	Data Sources	Simulator	Query Creation	Answer	Size
EQA v1 [138]	2018	Active EQA	SUNCG	House3D	Rule-Based	open-ended	5,000+
MT-EQA [139]	2019	Active EQA	SUNCG	House3D	Rule-Based	open-ended	19,000+
MP3D-EQA [140]	2019	Active EQA	MP3D	Simulator based on MINOS	Rule-Based	open-ended	1,136
IQUAD V1 [141]	2018	Interactive EQA	-	AI2THOR	Rule-Based	multi-choice	75,000+
VideoNavQA [142]	2019	Episodic Memory EQA	SUNCG	House3D	Rule-Based	open-ended	101,000
SQA3D [143]	2022	QA only	ScanNet	-	Manual	multi-choice	33,400
K-EQA [144]	2023	Active EQA	-	AI2THOR	Rule-Based	open-ended	60,000
OpenEQA [145]	2024	Active EQA, Episodic Memory EQA	ScanNet, HM3D	Habitat	Manual	open-ended	1,600+
HM-EQA [146]	2024	Active EQA	HM3D	Habitat	VLM	multi-choice	500
S-EQA [147]	2024	Active EQA	-	VirtualHome	LLM	binary	-
EXPRESS-Bench [148]	2025	Exploration-aware EQA	HM3D	Habitat	VLM	open-ended	2,044



Fig. 7. The gray box displays the scenes an agent observes during exploration. The other boxes show various types of question answering tasks. Except for the task of answering questions based on episodic memory, the agent ceases exploration once it has gathered sufficient information to answer the question.

to answer the given questions. An agent with autonomous exploration and decision-making capabilities must not only consider which actions to take to explore the environment but also determine when to stop exploring to answer questions. Existing works focus on different types of questions, some of which are shown in Fig. 7. In this section, we will introduce the existing datasets, discuss the related methods, describe the metrics used to evaluate model performance, and address the remaining limitations of this task.

1) *Datasets*: We briefly introduce several embodied question answering datasets, which are summarized in Table VI.

EQA v1 [138] is the first dataset designed for EQA. Built on synthetic 3D indoor scenes from the SUNCG dataset [73] within the House3D [149] simulator, EQA v1 comprises four types of questions: location, color, color_room, and preposition. Similar to EQA v1, **MT-EQA** [139] is built in House3D using SUNCG by executing functional programs consisting of some basic operations. However, it further extends the single-object question answering task to a multi-object setting. Six types of questions are designed, involving the comparison of color, distance, and size between multiple objects. **MP3D-EQA** [140] is built on a simulator developed based on MINOS [150] using the Matterport3D dataset [151], expanding the question-answering task to a realistic 3D environment. Referring to EQA v1, MP3D-EQA utilizes three types of templates: location, color, and color_room, generating a total of 1,136

questions in 83 home environments. **IQUAD V1** [141] is built upon AI2-THOR and consists of three types of questions: existence, counting, and spatial relationships. Unlike other datasets, answering IQUAD V1 questions requires the agent to have a good understanding of affordances and interact with the dynamic environment. **VideoNavQA** [142] decouples the visual reasoning from the navigation aspect of the EQA problem. In this task, the agent accesses videos corresponding to exploration trajectories with sufficient information to answer questions. **SQA3D** [143] simplifies protocol (QA only) while still preserving the function of benchmarking embodied scene understanding, enabling more complex, knowledge-intensive questions and a much larger scale of data collection.

Unlike previous datasets that explicitly specify target objects in questions, **K-EQA** [144] features complex questions with logical clauses and knowledge-related phrases, requiring prior knowledge to answer. **OpenEQA** [145] is the first open-vocabulary dataset for EQA, supporting both episodic memory and active exploration cases. The episodic memory EQA (EM-EQA) tasks involve an agent developing an understanding of the environment from its episodic memory to answer questions. In active EQA (A-EQA) tasks, the agent answers questions by taking exploratory actions to gather necessary information. Utilizing GPT4-V, **HM-EQA** [146] is constructed in the Habitat simulator using HM3D. It includes 500 questions across 267 different scenes, which can be roughly categorized into identification, counting, existence, status, and location. **S-EQA** [147] leverages GPT-4 in VirtualHome for data generation and employs cosine similarity calculations to decide whether to retain the generated data, thereby enhancing dataset diversity. In S-EQA, answering questions requires the assessment of a collection of consensus objects and states to reach an existential “Yes/No” answer. **EXPRESS-Bench** [148] is the largest exploration-aware EQA dataset that consists of 777 exploration trajectories and 2,044 samples. It also introduces novel evaluation metrics to ensure faithful assessment.

2) *Methods*: The embodied question answering task mainly involves navigation and question-answering subtasks, with implementation methods broadly categorized into two types: neural network-based and LLMs/VLMs-based.

Neural Network Methods. In early work, researchers mainly addressed the embodied question answering task by building deep neural networks. They trained and fine-tuned these models using techniques such as imitation learning and reinforcement learning to improve performance.

The EQA task was first proposed by Das et al. [138]. In their work, the agent consists of four main modules:

表VI
不同 EQA 数据集的比较

数据集	年	类型	数据源	模拟器	查询创建	答案	尺寸
EQA v1 [138]	2018	活动性 EQA	SUNCG	House3D	基于规则的	没有固定限度的	5,000+
MT-EQA [139]	2019	活动性 EQA	SUNCG	House3D	基于规则的	没有固定限度的	19,000+
MP3D-EQA [140]	2019	活动性 EQA	MP3D	基于minos的模拟器	基于规则的	没有固定限度的	1,136
IQUAD V1 [141]	2018	交互式 EQA	-	AI2THOR	基于规则的	多选	75,000+
VideoNavQA [142]	2019	情景记忆 EQA	SUNCG	House3D	基于规则的	没有固定限度的	101,000
SQA3D [143]	2022	仅限 QA 使用	扫描网	-	手动	多选	33,400
K-EQA [144]	2023	活动性 EQA	-	AI2THOR	基于规则的	没有固定限度的	60,000
OpenEQA [145]	2024	主动 EQA, 情景记忆 EQA	ScanNet, HM3D	栖息地	手动	没有固定限度的	1,600+
HM-EQA [146]	2024	活动性 EQA	HM3D	栖息地	VLM	多选	500
S-EQA [147]	2024	活动性 EQA	-	虚拟家园	LLM	二进制	-
Express-bench[148]	2025	探索感知 EQA	HM3D	栖息地	VLM	没有固定限度的	2,044



图 7. 灰色方框显示智能体在探索过程中观察到的场景；其他方框则展示了各类问答任务。除基于情景记忆进行问答的任务外，智能体在收集到足够信息以回答问题后便会停止探索。

为解答给定问题，具备自主探索与决策能力的智能体不仅要考虑采取何种行动来探索环境，还需判断何时停止探索以获取问题答案。现有研究聚焦于不同类型的问题，部分示例如图7所示。本节将介绍现有数据集、探讨相关方法、阐述模型性能评估指标，并分析该任务仍存在的局限性。

1) 数据集：我们简要介绍几个具身化问答数据集，其总结见表VI。**EQA v1**[138]是首个专为 EQA 设计的数据集。该数据集基于House3D[149]模拟器中来自 SUNCG 数据集[73]的合成3D室内场景构建，包含位置、颜色、彩色房间和介词四种类型的问题。与 EQA v1类似，**MT-EQA**[139]通过执行由基本操作组成的功能程序，在House3D中采用 SUNCG 实现。但其将单对象问答任务扩展至多对象场景，设计了六种涉及多个对象间颜色、距离和尺寸比较的问题类型。**MP3DEQA**[140]基于minos[150]开发的模拟器构建，利用Matterport3D数据集[151]将问答任务拓展至真实3D环境。与 EQA v1类似，MP3D-EQA 采用位置、颜色和彩色房间三种模板类型，在83种家居环境中生成共计1,136个问题。**IQUAD V1**[141]基于AI2-THOR构建，包

含存在性判断、数量统计和空间关系三种类型的问题。与其他数据集不同，回答 IQUAD V1问题需要智能体具备良好的可操作性理解能力，并能与动态环境进行交互。**VideoNavQA**[142]将视觉推理与 EQA 问题中的导航维度解耦。在该任务中，智能体通过访问与探索轨迹对应的视频来获取足够信息以回答问题。**SQA3D**[143]在简化协议（仅限问答功能）的同时，仍保留了基准测试具身场景理解能力的功能，从而支持更复杂的知识密集型问题并实现更大规模的数据采集。

与以往在问题中明确指定目标对象的数据集不同，**K-EQA**[144]处理的是包含逻辑子句和知识相关短语的复杂问题，需要先验知识才能解答。**OpenEQA**[145]是首个面向 EQA 的开放词汇数据集，同时支持情景记忆和主动探索两种任务类型。情景记忆 EQA（EMEQA）任务要求智能体通过情景记忆理解环境以回答问题；而主动 EQA（A-EQA）任务则要求智能体通过执行探索性动作来收集必要信息以解答问题。基于GPT-4-V，**HM-EQA**[146]是在Habitat模拟器中使用HM3D构建的，包含覆盖267个不同场景的500道题目，大致可分为识别、计数、存在性、状态和位置五大类。**S-EQA**[147]利用VirtualHome中的GPT-4进行数据生成，并采用余弦相似度计算来决定是否保留生成数据，从而提升数据集的多样性。在S-EQA中，回答问题需要评估一系列共识对象和状态，以得出存在性的“是/否”答案。**express-Bench**[148]是最大的具备探索感知能力的 EQA 数据集，包含777条探索轨迹和2,044个样本。该数据集还引入了新的评估指标，以确保评估结果的准确性。

2) 方法：该具身化问答任务主要包含导航与问答子任务，其实现方法可大致分为两类：基于神经网络的方法和基于大语言模型（LLMs）/视觉语言模型（VLMs）的方法。

神经网络方法。早期研究中，学者们主要通过构建深度神经网络来解决具身化问答任务。他们采用模仿学习和强化学习等技术对模型进行训练与微调，从而提升性能表现。

EQA 任务最初由Das等人[138]提出。在他们的研究中，智能体由四个主要模块组成：

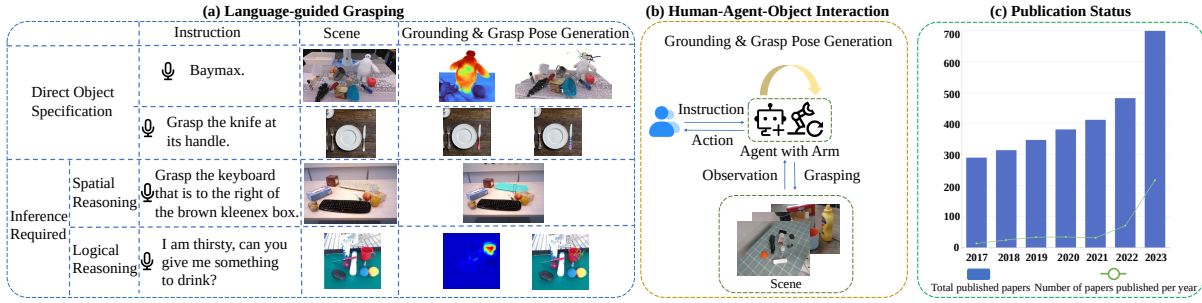


Fig. 8. The overview of the embodied grasping task. (a) demonstrates examples of language-guided grasping for different types of tasks, (b) provides an overview of human-agent-object interaction, (c) shows Google Scholar search results for topics of “Language-guided Grasping”.

vision, language, navigation, and answering. These modules are primarily constructed using traditional neural building blocks: Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). Some subsequent works [152] retained modules like the question answering module proposed by Das et al. [138] and improved the model. Additionally, Wu et al. [152] proposed integrating the navigation and QA modules into a unified SGD training pipeline for joint training, thereby avoiding employing deep reinforcement learning to simultaneously train the separately trained navigation and question answering modules. From the perspective of task singularity, several works [153] expanded the task to include multiple objectives and multi-agent, respectively, making it necessary for the model to store and integrate the information obtained by the agent’s exploration through methods such as feature extraction and scene reconstruction. Considering the interaction between the agent and the dynamic environment, Gordon et al. [141] introduced the Hierarchical Interactive Memory Network. There is also a limitation in previous works where agents are unable to use external knowledge to answer complex questions and lack knowledge of the explored parts of the scene. To address this, Tan et al. [144] leveraged the neural program synthesis method and the table converted from the knowledge and 3D scene graphs, allowing the action planner to access object-related information. Additionally, an approach based on Monte Carlo Tree Search (MCTS) is used to determine the next location for the agent.

LLMs/VLMs Methods. Majumdar et al. [145] used LLMs and VLMs for episodic memory EQA (EM-EQA) and Active EQA (A-EQA) tasks. For EM-EQA task, they considered Blind LLMs, Socratic LLMs with language descriptions of the episodic memory, Socratic LLMs with descriptions of the constructed scene graph, and VLMs processing multiple scene frames. The A-EQA task extended EM-EQA methods with frontier-based exploration (FBE) [154] for problem-independent environment exploration. Some works [146], [155] employed frontier-based exploration method to identify areas for subsequent exploration and to build semantic maps. They ended the exploration early utilizing conformal prediction or image-text matching to avoid over-exploration. Patel et al. [156] emphasized the question answering aspect of the task. They leveraged multiple LLM-based agents to explore the environment and enable them to independently answer questions with “yes” or “no” answers. These individual responses are utilized to train a Central Answer Model, to aggregate responses and generate robust answers.

3) *Limitations:* a) Dataset: Constructing datasets requires substantial manpower and resources. Additionally, there are still few large-scale datasets, and the metrics for evaluating model performance vary across different datasets, complicating the testing and comparison of performance, b) Model: Despite the advancements brought by LLMs, the performance of these models still lags significantly behind human levels. Future work may focus more on effectively storing environmental information explored by agents and guiding them to plan actions based on environmental memory and questions, while also enhancing the interpretability of their actions.

B. Embodied Grasping

Embodied interaction includes not only question-answering but also performing tasks like grasping and placing objects based on human instructions. Embodied grasping combines traditional kinematic methods [157], [158] with large models such as LLMs and vision-language models, enabling multi-sensory perception and reasoning for task execution. Figure 8 (b) shows an overview of human-agent-object interactions where embodied grasping is performed.

1) *Datasets:* Recently, a substantial number of grasping datasets [159]–[163] have been generated. These datasets typically contain annotated grasping data based on images (RGB, depth), point clouds, or 3D scenes. With the advent of MLMs and the application of foundational language models to robotic grasping, there is an urgent need for datasets that include linguistic text. Consequently, existing datasets have been extended or reconstructed to create semantic-grasping datasets [165]–[167]. These datasets are instrumental in studying grasping models grounded in language, enabling agents to develop a broad understanding of semantics.

Traditional grasping datasets encompass data for both single objects [159] and cluttered scenes [164], providing stable grasp annotations (4-DOF or 6-DOF) that conform to kinematics for each object. These data can be collected from real desktop environments [159], typically including RGB, depth, and point cloud data, or from virtual environments [162], which include image data, point clouds, or scene models. While these datasets are useful for grasping models, they lack semantic information. To bridge this gap, these datasets have been augmented or extended with semantic expressions [165], [168], thereby linking language, vision, and grasping. By incorporating semantic information, agents can better understand and execute grasping tasks. This enhancement allows for the

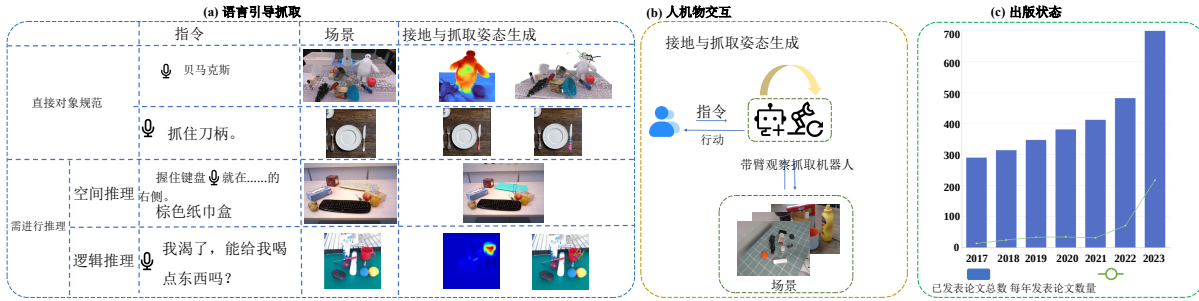


图 8. 内在化抓取任务的概述。(a) 展示了不同类型任务中语言引导抓取的示例；(b) 概述了人-智能体-物体交互过程；(c) 显示了关于“语言引导抓取”主题的 Google Scholar 搜索结果。

视觉、语言、导航和问答功能。这些模块主要采用传统的神经网络构建模块实现：卷积神经网络（CNN）和循环神经网络（RNN）。后续部分研究[152]保留了Das等人[138]提出的问答模块等组件，并对模型进行了改进。此外，Wu等人[152]提出将导航与问答模块整合到统一的SGD训练流程中进行联合训练，从而避免使用深度强化学习同时训练分别训练的导航和问答模块。从任务单一性角度出发，多项研究[153]分别扩展了任务以包含多目标和多智能体场景，这要求模型通过特征提取和场景重建等方法存储并整合智能体探索过程中获取的信息。考虑到智能体与动态环境之间的交互，Gordon等人[141]提出了分层交互记忆网络。现有研究还存在一个局限性：智能体无法利用外部知识解答复杂问题，且对场景中已探索区域缺乏认知。为解决这一问题，Tan等人[144]采用神经程序合成方法，并通过知识库与三维场景图转换生成的表格，使动作规划器能够获取物体相关数据。此外，研究还运用基于蒙特卡罗树搜索（MCTS）的方法来确定智能体的下一步行动位置。

LLM/VLM方法。Majumdar等人[145]将LLMs和VLM应用于情景记忆EQA（EM-EQA）和主动EQA（A-EQA）任务。在EM-EQA任务中，他们采用了盲向LLMs、结合情景记忆语言描述的苏格拉底式LLMs、结合构建场景图描述的苏格拉底式LLMs，以及处理多场景帧的VLM。A-EQA任务在EM-EQA方法基础上，引入了基于前沿的探索（FBE）[154]技术，用于进行与问题无关的环境探索。部分研究[146]、[155]采用基于前沿的探索方法来识别后续探索区域并构建语义地图，随后通过一致性预测或图像-文本匹配提前结束探索以避免过度探索。Patel等人[156]重点强调了该任务的问题回答功能：他们利用多个基于LLM的智能体进行环境探索，并使其能够独立给出“是”或“否”的回答；这些独立回答被用于训练中央答案模型，从而聚合回答并生成稳健的答案。

3) **局限性：**a) 数据集：构建数据集需要投入大量人力和资源。此外，目前仍缺乏大规模数据集，且不同数据集间评估模型性能的指标存在差异，这给性能测试与对比带来了复杂性；b) 模型：尽管大语言模型（LLM）带来了显著进步，但其性能仍远低于人类水平。未来研究可重点聚焦于有效存储智能体探索的环境信息，并基于环境记忆与问题引导其制定行动方案，同时提升行为的可解释性。

B. 具身化抓取

具身交互不仅包含问答功能，还涉及根据人类指令执行抓取、放置物体等任务。具身抓取技术将传统运动学方法[157]、[158]与大模型（如语言模型和视觉语言模型）相结合，实现了任务执行过程中的多感官感知与推理能力。图8(b)展示了具身抓取技术在人-智能体-物体交互场景中的应用示意图。

1) **数据集：**近年来，大量抓取数据集[159]–[163]已被生成。这些数据集通常包含基于图像（RGB、深度信息）、点云或三维场景标注的抓取数据。随着多语言模型（MLM）的出现及基础语言模型在机器人抓取领域的应用，亟需包含语言文本的数据集。因此，现有数据集经过扩展或重构后形成了语义抓取数据集[165]–[167]。这类数据集对于研究基于语言的抓取模型具有关键作用，使智能体能够建立对语义的广泛理解。

传统抓取数据集包含单个物体[159]与复杂场景[164]的数据，提供符合各物体运动学特性的稳定抓取标注（4 DOF 或 6 DOF）。这些数据可采集自真实桌面环境[159]（通常包含RGB图像、深度数据及点云数据），或虚拟环境[162]（包含图像数据、点云或场景模型）。尽管此类数据集对抓取模型具有重要价值，但缺乏语义信息。为弥补这一缺陷，研究者通过添加语义表达式对数据集进行增强[165]、[168]，从而实现语言、视觉与抓取的融合。通过整合语义信息，智能体能够更精准理解并执行抓取任务。这种增强技术使得

TABLE VII
EMBODIED GRASPING DATASETS.

Dataset	Year	Type	Modality	Grasp Label	Gripper Finger	Objects	Grasps	Scenes	Language
Cornell [159]	2011	Real	RGB-D	Rect.	2	240	8K	Single	×
Jacquard [160]	2018	Sim	RGB-D	Rect.	2	11K	1.1M	Single	×
6-DOF GraspNet [161]	2019	Sim	3D	6D	2	206	7.07M	Single	×
ACRONYM [162]	2021	Sim	3D	6D	2	8872	17.7M	Multi	×
MultiGripperGrasp [163]	2024	Sim	3D	-	2-5	345	30.4M	Single	×
OCID-Grasp [164]	2021	Real	RGB-D	Rect.	2	89	75K	Multi	×
OCID-VLG [165]	2023	Real	RGB-D,3D	Rect.	2	89	75K	Multi	✓
ReasoningGrasp [166]	2024	Real	RGB-D	6D	2	64	99.3M	Multi	✓
CapGrasp [167]	2024	Sim	3D	-	5	1.8K	50K	Single	✓

development of more sophisticated and semantically aware grasping models, facilitating more intuitive and effective interaction with the environment. Table VII presents the datasets described above, including traditional grasping datasets and language-based grasping datasets.

2) *Language-guided grasping*: The concept of language-guided grasping [165], [166], [168], which has evolved from this integration, combines MLMs to provide agents with the capability of semantic scene reasoning. This allows the agent to execute grasping operations based on implicit or explicit human instructions. Figure 8 (c) illustrates the publication trends in recent years on the topic of language-guided grasping. With the advancement of LLMs, researchers have shown increasing interest in this topic. Currently, grasping research is increasingly focused on open-world scenarios, emphasizing the open-set generalization [169] methods. By leveraging the generalization capabilities of MLMs, robots can perform grasping tasks in open-world environments with greater intelligence and efficiency.

In language-guided grasping, semantics can originate from explicit instructions [169], [170] and implicit instructions [166], [167]. Explicit instructions clearly specify the category of the object to be grasped, such as a banana or an apple. Implicit instructions, however, require reasoning to identify the object or a part of the object to be grasped, involving spatial reasoning and logical reasoning.

Spatial reasoning [165] refers to instructions that may include the spatial relationship of the object or part to be grasped, necessitating the inference of grasping posture based on the spatial relationships of objects within the scene. For example, “Grasp the keyboard that is to the right of the brown kleenex box” involves understanding and inferring the spatial arrangement of objects. Logical reasoning [166], on the other hand, involves instructions that may contain logical relationships requiring inference to discern human intent and subsequently grasp the target. For instance, “I am thirsty, can you give me something to drink?” would prompt the agent to potentially hand over a glass of water or a bottle of a beverage. The agent must ensure that the liquid does not spill during the handover, thus generating a reasonable grasping posture.

In both cases, the integration of semantic understanding with spatial and logical reasoning enables the agent to perform complex grasping tasks effectively and accurately. Figure 8 (a) depicts various types of language-guided grasping tasks.

3) *End-to-End Approaches*: CLIPORT [168] is a language-conditioned imitation learning agent that combines the vision-language pre-trained model CLIP with the Transporter Net to create an end-to-end dual-stream architecture for semantic understanding and grasp generation. It is trained using a large

number of expert demonstration data collected from virtual environments, enabling the agent to perform semantically guided grasping. Based on the OCID dataset, CROG [165] proposes a vision-language-grasping dataset and introduces a competitive end-to-end baseline. It leverages CLIP’s visual foundation capabilities to learn grasp synthesis directly from image-text pairs. Reasoning Grasping [166] introduces the first reasoning grasping benchmark dataset based on the GraspNet-1 Billion dataset and proposes an end-to-end reasoning grasping model. The model integrates multimodal LLMs with vision-based robotic grasping frameworks to generate grasps based on semantics and vision. SemGrasp [167] is a method for semantic-based grasp generation that incorporates semantic information into grasp representations to generate dexterous hand grasp postures. It introduces a discrete representation aligning grasp space with semantic space, enabling the generation of grasp postures according to language instructions.

4) *Modular Approaches*: F3RM [169] seeks to elevate CLIP’s text-image priors into 3D space, using extracted features for language localization followed by grasp generation. It combines precise 3D geometry with rich semantics from 2D foundational models, utilizing features extracted from CLIP to specify objects for manipulation through free-text natural language. It demonstrates the ability to generalize to unseen expressions and new object categories. GaussianGrasper [170] utilizes a 3D Gaussian field to achieve language-guided grasping tasks. The proposed methodology begins with the construction of a 3D Gaussian field, followed by feature distillation. Subsequently, language-based localization is performed using the extracted features. Finally, grasp pose generation is carried out based on a SOTA pre-trained grasping network [171]. It integrates open-vocabulary semantics with precise geometry, enabling grasping based on language instructions.

These approaches advance language-guided grasping by using end-to-end and modular frameworks, enhancing robotic agents’ ability to perform complex grasping tasks from natural language instructions. Embodied grasping improves robots’ intelligence and utility in-home services and industrial manufacturing. However, current methods face limitations, including reliance on extensive data and poor generalization. Future research aims to enhance agent generality, enabling robots to understand complex semantics, grasp a variety of unseen objects, and tackle intricate tasks.

VI. EMBODIED AGENT

An agent is defined as an autonomous entity capable of perceiving its environment and acting to achieve specific objectives. Recent advancements in MLMs have further expanded

表VII呈现了数据集抓取结果。

数据集	年	类型	模态	抓握标签	夹持指	物体	抓握	场景	语言
康奈尔大学[159]	2011	真的	RGB-D	矩形。	2	240	8K	单人	×
提花织机[160]	2018	同时	RGB-D	直角	2	11K	1.1M	单人	×
6-DOF GraspNet [161]	2019	同时	3D	6D	2	206	7.07M	单人	×
缩写词[162]	2021	同时	3D	6D	2	8872	17.7M	多个: 多种	×
MultiGripperGrasp [163]	2024	同时	3D	-	2-5	345	30.4M	单人	×
OCID-Grasp [164]	2021	真的	RGB-D	矩形。	2	89	75K	多个: 多种	×
OCID-VLG [165]	2023	真的	RGB-D,3D	矩形。	2	89	75K	多个: 多种	✓
ReasoningGrasp [166]	2024	真的	RGB-D	6D	2	64	99.3M	多个: 多种	✓
CapGrasp [167]	2024	同时	3D	-	5	1.8K	50K	单人	✓

开发更复杂且具有语义感知能力的抓取模型，以实现与环境更直观、高效的交互。表VII列出了上述数据集，包括传统抓取数据集和基于语言的抓取数据集。

2) *语言引导抓取*: 语言引导抓取概念[165]、[166]、[168]源于该融合技术发展而来，通过整合多语言模型（MLMs）赋予智能体语义场景推理能力。这使得智能体能够根据显性或隐性人类指令执行抓取操作。图8(c)展示了近年来语言引导抓取领域的论文发表趋势。随着大语言模型（LLMs）的持续发展，研究者对该领域的关注度与日俱增。当前抓取研究正日益聚焦于开放世界场景，重点强调开放集泛化[169]方法。借助MLMs的泛化能力，机器人在开放世界环境中可实现更高智能水平与执行效率的抓取任务。

在语言引导的抓取过程中，语义信息可来源于明确指令[169]、[170]以及隐性指令[166]、[167]。明确指令会清晰指定待抓取物体的类别（例如香蕉或苹果）；而隐性指令则需要通过推理来识别待抓取物体或其组成部分，这一过程涉及空间推理与逻辑推理。

空间推理[165]是指需要理解被抓取物体或部件的空间关系，并根据场景中物体的空间布局推断抓取姿势的指令类型。例如“抓取棕色纸巾盒右侧的键盘”这类指令，就需要理解并推断物体的空间排列关系。而逻辑推理[166]则涉及包含逻辑关系的指令，需要通过推理来判断人类意图并完成抓取动作。比如“我口渴了，能给我喝点东西吗？”这类指令会促使智能体选择递送水杯或饮料瓶，同时必须确保递送过程中液体不洒漏，从而形成合理的抓取姿势。

在这两种情况下，语义理解与空间及逻辑推理的整合使智能体能够高效且准确地执行复杂的抓取任务。图8(a)展示了多种语言引导型抓取任务的类型。

3) *端到端方法*: cliport[168]是一种语言条件模仿学习智能体，它将视觉语言预训练模型CLIP与Transporter Net相结合，构建了用于语义理解与抓取生成的端到端双流架构。该模型通过从虚拟环境中收集的大量专家演示数据进

行训练，使智能体能够执行语义引导式抓取操作。基于OCID数据集，CROG[165]提出了视觉-语言-抓取数据集，并引入了具有竞争力的端到端基线模型。该模型利用CLIP的视觉基础能力，直接从图像-文本对中学习抓取合成。Reasoning Grasping[166]首次基于GraspNet-1 Billion数据集提出推理抓取基准数据集，并开发了端到端推理抓取模型。该模型将多模态大语言模型与基于视觉的机器人抓取框架相结合，能够根据语义信息和视觉特征生成抓取动作。SemGrasp[167]是一种基于语义的抓取生成方法，该方法将语义信息融入抓取表征中，以生成灵巧的手部抓取姿势。该方法引入了一种离散表征，将抓取空间与语义空间对齐，从而能够根据语言指令生成相应的抓取姿势。

4) *模块化方法*: F3RM[169]致力于将CLIP的文本-图像先验信息提升至三维空间，利用提取的特征进行语言定位并随后生成抓取动作。该方法结合精确的三维几何信息与来自二维基础模型的丰富语义，通过从CLIP提取的特征，利用自由文本自然语言指定待操作对象。实验表明其能够泛化到未见过的表情和新的物体类别。GaussianGrasper[170]采用三维高斯场实现语言引导的抓取任务：首先构建三维高斯场，随后进行特征蒸馏；接着利用提取的特征进行基于语言的定位；最后基于SOTA预训练抓取网络[171]生成抓取姿态。该方法将开放词汇语义与精确几何信息相结合，实现了基于语言指令的抓取操作。

这些方法通过采用端到端和模块化框架，推动了基于语言的抓取技术发展，显著提升了机器人根据自然语言指令执行复杂抓取任务的能力。具身化抓取技术增强了机器人在家庭服务和工业制造领域的智能水平与实用性。然而，现有方法仍存在局限性，例如依赖海量数据且泛化能力较弱。未来的研究目标是提升机器人的通用性，使其能够理解复杂的语义信息、抓取多种未见过的物体，并应对复杂任务。

VI. 有体智能体

智能体被定义为能够感知环境并采取行动以实现特定目标的自主实体。多层次营销（MLM）领域的最新进展进一步拓展了该概念的应用范围。

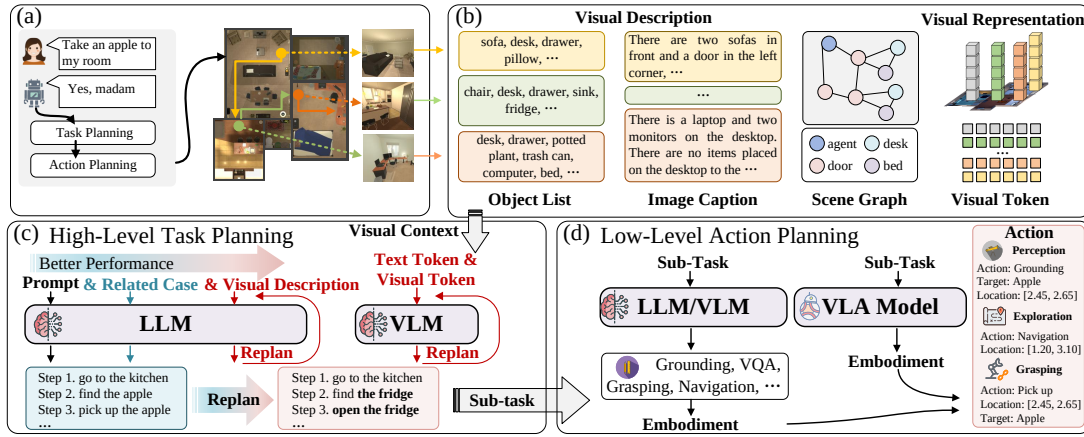


Fig. 9. The architecture of the embodied agent based on embodied multimodal foundation model, which consists of visual perception module, high-level task planning module, and low-level action planning module.

the application of agents to practical scenarios. When these MLM-based agents are embodied in physical entities, they can effectively transfer their capabilities from virtual space to physical world, thereby becoming embodied agents [172].

To enable embodied agents to operate in the information-rich and complex real world, the embodied agents have been developed to show strong multimodal perception, interaction and planning capabilities, as shown in Fig. 9. To complete a task, embodied agents typically involves the following process: 1) decomposing the abstract and complex task into specific subtasks, which is referred to as high-level Embodied Task Planning. 2) gradually implementing these subtasks by effectively utilizing Embodied Perception and Embodied Interaction models or leveraging the Foundation Model's policy function, named low-level Embodied Action Planning. It is worth noting that task planning involves thinking before acting, and is therefore typically considered in cyber space. In contrast, action planning must account for effective interaction with the environment and feedback on this information to the task planner to adjust task planning. Thus, it is crucial for embodied agents to align and generalize their abilities from the cyber space to the physical world.

A. Embodied Task Planning

Traditional embodied task planning methods are usually based on explicit rules and logical reasoning. For example, symbolic planning algorithms such as PDDL [173], and search algorithms like MCTS [174] and A* [175], are used to generate plans. However, these methods often rely on predefined rules, constraints, and heuristics that are rigid and may not adapt well to dynamic or unforeseen changes. With the popularity of LLMs, many works have attempted to use LLMs for planning or to combine traditional methods with LLMs, leveraging the rich embedded world knowledge for reasoning and planning without the need for handcrafted definitions, greatly enhancing the model's generalization capabilities.

1) Planning utilizing the Emergent Capabilities of LLMs:

Before the scale-up of natural language models, task planners were similarly implemented by training models like BERT on embodied instruction datasets such as Alfred [176] and Alfworld [177], as demonstrated by FILM [178]. However,

this approach was limited by the examples in the training set and could not effectively align with the physical world. Nowadays, thanks to the emergent capabilities of LLMs, they can decompose abstract tasks using their internal world knowledge and chain-of-thought reasoning, similar to how humans reason through task completion steps before acting. For example, Translated LM [179] and Inner Monologue [180] can break down complex tasks into manageable steps and devise solutions using their internal logic and knowledge systems without additional training. Similarly, the multi-agent collaboration framework ReAd [181] efficiently self-refined plans via different prompts. Additionally, some approaches abstract past successful examples into a series of skills stored in a memory bank to consider during inference and improve planning success rates [182]–[184]. Some works utilized code as the reasoning medium instead of natural language, where task planning is generated as code based on the available API library [185], [186]. Furthermore, multi-turn reasoning can effectively correct potential hallucinations in task planning. For instance, Socratic Models [187] and Socratic Planner [188] used Socratic questioning to derive reliable planning.

However, during task planning, potential failures may occur during execution, often resulting from the planner not fully accounting for the complexity of the real environment and the difficulty of task execution [180], [189]. Due to a lack of visual information, planned subtasks may deviate from the actual scenario, leading to task failure. Therefore, integrating visual information into planning or replanning during execution is necessary. This approach can significantly enhance the accuracy and feasibility of task planning, better addressing the challenges of real-world environments.

2) *Planning utilizing the visual information from embodied perception model:* Based on the above discussion, it is important to integrate visual information into task planning (or replanning). In this process, object labels, locations, or descriptions provided by visual input can offer critical references for task decomposition and execution by LLMs. Through visual information, LLMs can more accurately identify target objects and obstacles in the current environment, thereby optimizing task steps or modifying subtask objectives. Some works use an object detector to query the objects present in the

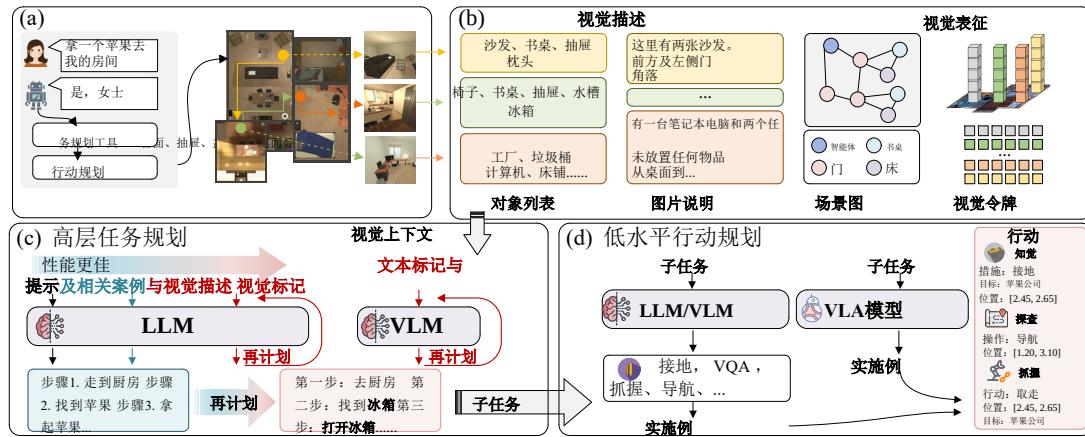


图9. 基于具身多模态基础模型的具身智能体架构, 该架构由视觉感知模块、高层任务规划模块及低层动作规划模块组成。

智能体在实际场景中的应用。当这些基于 MLM 的智能体被具象化为物理实体时, 它们能够将自身能力从虚拟空间有效迁移至物理世界, 从而成为具身智能体[172]。

为了让具身智能体能够在信息密集且复杂的现实世界中运行, 研究人员开发了具备强大多模态感知、交互及规划能力的具身智能体, 如图9所示。完成任务通常需要经历以下流程: 1) 将抽象复杂的任务分解为具体子任务, 这被称为高层级具身任务规划; 2) 通过有效运用具身感知模型、具身交互模型或利用基础模型的策略函数逐步实施这些子任务, 即低层级具身动作规划。需要特别指出的是, 任务规划本质上属于“先思考后行动”的思维模式, 因此通常应用于网络空间场景。相比之下, 动作规划必须考虑与环境的有效交互, 并将反馈信息及时传递给任务规划系统以进行动态调整。因此, 具身智能体必须确保其能力在网络空间与物理世界之间保持一致性并具备迁移能力。

A. 具身任务规划

传统的嵌入式任务规划方法通常基于显式规则和逻辑推理。例如, PDDL 等符号规划算法[173], 以及 MCTS[174]和A*算法[175]等搜索算法常被用于生成规划方案。然而这些方法往往依赖预定义的规则、约束条件和启发式算法, 存在固有局限性, 难以有效应对动态变化或突发状况。随着大语言模型(LLMs)的广泛应用, 众多研究尝试将LLMs应用于规划领域, 或将传统方法与LLMs相结合——通过利用丰富的嵌入式世界知识进行推理规划, 无需人工定义参数, 从而显著提升模型的泛化能力。

1) *利用大语言模型涌现能力进行规划*: 在自然语言模型规模化应用之前, 任务规划器的实现方式类似——通过在BERT等模型上使用Alfred[176]和Alfworld[177]等具身指令数据集进行训练, 正如FILM[178]所展示的那样。然而

这种方法受限于训练集中的示例数量, 难以有效与物理世界对齐。如今得益于大语言模型的涌现能力, 它们能够利用内部世界知识和思维链推理来分解抽象任务, 类似于人类在行动前通过任务完成步骤进行推理的方式。例如, Translated LM[179]和Inner Monologue[180]无需额外训练即可将复杂任务分解为可管理步骤, 并运用内部逻辑和知识系统设计解决方案。类似地, 多智能体协作框架ReAd[181]能通过不同提示高效实现计划的自我优化。此外, 部分方法将过往成功案例抽象为存储在记忆库中的系列技能, 在推理过程中加以考量以提升规划成功率[182]–[184]。另有研究采用代码作为推理媒介替代自然语言, 通过现有API库将任务规划生成代码形式[185], [186]。此外, 多轮推理能有效修正任务规划中的潜在幻觉。例如, 苏格拉底模型[187]与苏格拉底规划器[188]通过采用苏格拉底式提问法来推导可靠规划方案。

然而在任务规划过程中, 执行阶段可能出现潜在失败, 这通常源于规划者未能充分考虑真实环境的复杂性及任务执行难度[180]、[189]。由于缺乏视觉信息支持, 预先规划的子任务可能与实际场景存在偏差, 从而导致任务失败。因此, 在执行阶段将视觉信息整合到规划或重新规划过程中具有必要性。该方法能显著提升任务规划的准确性和可行性, 更有效地应对现实环境中的挑战。

2) *基于嵌入式感知模型的视觉信息规划方法*: 综上所述, 将视觉信息融入任务规划(或重新规划)过程至关重要。在此过程中, 视觉输入提供的物体标签、位置信息或描述可为大语言模型(LLMs)的任务分解与执行提供关键参考。通过视觉信息, LLMs能更精准地识别当前环境中的目标物体与障碍物, 从而优化任务步骤或调整子任务目标。部分研究采用物体检测器来查询环境中存在的物体

environment during task execution and feed this information back to the LLM, allowing it to modify unreasonable steps in the current plan [187], [189], [190]. RoboGPT considers the different names of similar objects within the same task, further improving the feasibility of replanning [8]. However, the information provided by labels is still too limited. Can further scene information be provided? SayPlan [191] proposes using hierarchical 3D scene graphs to represent the environment, effectively mitigating the challenges of task planning in large, multi-floor, and multi-room settings. Similarly, ConceptGraphs [192] also adopts 3D scene graphs to provide environmental information to LLMs. Compared to SayPlan, it offers more detailed open-world object detection and presents task planning in a code-based format, which is more efficient and better suited to the demands of complex tasks.

However, limited visual information can result in an agent's inadequate understanding of its environment. While LLMs are provided with visual cues, they often fail to capture the environment's complexity and dynamic changes, leading to misunderstandings and task failures. For example, if a towel is locked in a bathroom cabinet, the agent might repeatedly search the bathroom without considering this possibility [8]. To address this, more robust algorithms must be developed to integrate multiple sensory data, enhancing the agent's environmental understanding. Additionally, leveraging historical data and contextual reasoning, even when visual information is limited, can aid the agent in making reasonable judgments and decisions. This approach of multimodal integration and context-based reasoning not only increases task execution success rates but also provides new perspectives for the advancement of embodied artificial intelligence.

3) *Planning utilizing the VLMs*: Compared to converting environmental information into text using external visual models, VLM models can capture visual details in latent space, particularly contextual information that is difficult to represent with object labels. VLMs can discern rules underlying visual phenomena; for instance, even if a towel is not visible in the environment, it can be inferred that the towel might be stored in a cabinet. This process essentially demonstrates how abstract visual features and structured textual features can be more effectively aligned in latent space. In EmbodiedGPT [193], the Embodied-Former module aligns embodied, visual, and textual information, effectively considering the agent's state and environmental information during task planning. Unlike EmbodiedGPT, which directly uses third-person perspective images, LEO [194] encodes 2D egocentric images and 3D scenes into visual tokens. This method effectively perceives 3D world information and executes tasks accordingly. Similarly, the EIF-Unknow model utilizes Semantic Feature Maps extracted from Voxel Features as visual tokens, which are input along with text tokens into a trained LLaVA model for task planning [195]. Furthermore, embodied multimodal foundation models, or VLA models, have been extensively trained with large datasets in studies like the RT series [2], [9], PaLM-E [196], and Matcha [197] to achieve alignment of visual and textual features in embodied scenarios.

However, task planning is only the first step for an agent in completing an instruction task. Subsequent action planning

determines whether the task can be accomplished. In the experiments from RoboGPT [8], the accuracy of task planning reached 96%, but the overall task completion rate was only 60%, limited by the performance of the low-level planner. Therefore, whether an embodied agent can transition from the cyber space of "imagining how tasks are completed" to the physical world of "interacting with the environment and completing tasks" hinges on effective action planning.

B. Embodied Action Planning

The distinction between task planning and action planning highlights that action planning must address real-world uncertainties due to the insufficient granularity of task planning subtasks [198]. Action planning can be achieved by: 1) using pre-trained embodied models to complete subtasks via APIs, or 2) leveraging the VLA model's capabilities. The results from action planning are fed back to refine task planning.

1) *Action utilizing APIs*: A common approach involves providing LLMs with definitions of well-trained policy models to understand and use them effectively for specific tasks [189], [199]. By generating code, LLMs can abstract tools into a function library, allowing better handling of sub-tasks [186]. Reflexion adjusts these tools during execution to improve generalization [200]. DEPS enables LLMs to learn and combine various skills through zero-shot learning [201]. The hierarchical planning paradigm simplifies development by separating high-level task planning from specific action execution through policy models. This modularity allows independent development, testing, and optimization, enhancing flexibility and maintainability. While it enables adaptability to various tasks and environments, reliance on external policy models can introduce latency and affect performance, making the quality of these models crucial for overall agent effectiveness.

2) *Action utilizing VLA model*: Different from previous approach that task planning and action execution are performed within the same system, this paradigm leverages the capabilities of embodied multimodal foundation models for planning and executing actions, reducing communication latency and improving system response speed and efficiency. In VLA models, the tight integration of perception, decision-making, and execution modules allows the system to handle complex tasks and adapt to changes in dynamic environments more efficiently. This integration also facilitates real-time feedback, enabling the agent to self-adjust strategies, thereby enhancing the robustness and adaptability of task execution [10], [193], [202]. However, this paradigm is undoubtedly more complex and costly, particularly when dealing with intricate or long-term tasks. Additionally, a key issue is that an action planner, without an embodied world model, cannot simulate physical laws using only the internal knowledge of an LLM. This limitation hinders the agent to accurately and effectively complete various tasks in the physical world, preventing the seamless transfer from cyber space to physical world.

3) *Scalability in Diverse Environments*: Scalability in embodied agents involves adapting to increased complexity in larger and more diverse environments through robust perception, efficient decision-making, and resource optimization.

在任务执行过程中收集环境信息并反馈给大语言模型（LLM），使其能够修正当前计划中不合理步骤[187]、[189]、[190]。RoboGPT通过识别同一任务中相似物体的不同名称，进一步提升了重新规划的可行性[8]。然而标签提供的信息仍显不足，能否补充更多场景信息？SayPlan[191]提出采用分层3D场景图表示环境，有效缓解了大型多楼层、多房间场景下的任务规划难题。类似地，ConceptGraphs[192]也采用3D场景图向LLM提供环境信息。相较于SayPlan，该方案不仅实现了更精细的开放世界物体检测，还以代码化形式呈现任务规划方案，这种高效模式更能满足复杂任务需求。

然而，有限的视觉信息可能导致智能体对其所处环境的理解不足。尽管大型语言模型（LLM）能够接收视觉线索，但往往难以捕捉环境的复杂性与动态变化，从而引发认知偏差和任务失败。例如，当毛巾卡在浴室柜中时，智能体可能会反复搜索浴室区域而忽视这种可能性[8]。为解决这一问题，必须开发更强大的算法来整合多源感官数据，从而提升智能体的环境感知能力。此外，通过利用历史数据和上下文推理机制，即便在视觉信息匮乏的情况下，也能帮助智能体做出合理判断与决策。这种多模态整合与基于上下文的推理方法，不仅显著提高了任务执行成功率，更为具身人工智能的发展提供了全新视角。

3) *基于视觉语言模型的规划方法*：与通过外部视觉模型将环境信息转化为文本不同，VLM模型能够在潜在空间中捕捉视觉细节，尤其是难以用对象标签表示的上下文信息。视觉语言模型能够识别视觉现象背后的规律性特征——例如即使毛巾未在环境中可见，也能推断其可能存放在橱柜中。这一过程充分展示了如何在潜在空间中更有效地整合抽象视觉特征与结构化文本特征。在EmbodiedGPT[193]中，Embodied-Former模块通过整合具身化视觉信息与文本信息，在任务规划过程中有效考量智能体状态与环境信息。与直接使用第三人称视角图像的EmbodiedGPT不同，LEO[194]将二维自我中心图像与三维场景编码为视觉标记，该方法能有效感知三维世界信息并据此执行任务。类似地，EIF-Unknown模型利用从体素特征中提取的语义特征图作为视觉标记，这些标记与文本标记一同输入经过训练的LLaVA模型进行任务规划[195]。此外，具身化多模态基础模型（即VLA模型）已在RT系列研究[2]、[9]、PaLM-E[196]及Matcha[197]等研究中通过大规模数据集进行广泛训练，以实现具身化场景中视觉与文本特征的对齐。

然而，任务规划仅是智能体完成指令任务的第一步。后续的动作规划决定了任务能否成功完成。在RoboGPT[8]

的实验中，任务规划的准确率达到96%，但整体任务完成率仅为60%，这受限于底层规划器的性能表现。因此，具身智能体能否从“想象任务如何完成”的虚拟空间，过渡到“与环境互动并完成任务”的物理世界，关键在于有效的动作规划能力。

B. 具身行动规划

任务规划与行动规划之间的区别在于，行动规划必须应对现实世界中的不确定性——这源于任务规划子任务粒度不足的局限性[198]。行动规划可通过两种方式实现：1) 利用预训练具身模型通过API完成子任务；2) 充分发挥VLA模型的处理能力。行动规划产生的结果将反馈至任务规划系统进行优化调整。

1) *基于API的动作执行*：常见方法是向大语言模型提供经过充分训练的策略模型定义，使其能够有效理解和运用特定任务[189]、[199]。通过代码生成机制，大语言模型可将工具抽象为函数库，从而更高效地处理子任务[186]。反射机制在执行过程中动态调整这些工具以提升泛化能力[200]。DEPS技术使大语言模型能够通过零样本学习掌握并整合多种技能[201]。分层规划范式通过将高层任务规划与具体动作执行分离至策略模型层面，简化了开发流程。这种模块化设计支持独立开发、测试与优化，显著提升系统灵活性和可维护性。尽管该架构能适应不同任务场景，但对外部策略模型的依赖可能引入延迟并影响性能表现，因此模型质量对智能体整体效能具有决定性影响。

2) *基于VLA模型的动作执行*：与传统方法不同，该范式将任务规划与动作执行整合于同一系统中，充分利用具身化多模态基础模型的能力来实现动作规划与执行，从而降低通信延迟并提升系统响应速度与效率。在VLA模型中，感知模块、决策模块与执行模块的紧密集成使系统能够高效处理复杂任务，并更灵活地适应动态环境变化。这种集成还支持实时反馈机制，使智能体可自主调整策略，从而增强任务执行的鲁棒性与适应性[10]、[193]、[202]。然而该范式无疑更为复杂且成本高昂，尤其在处理复杂或长期任务时更为明显。此外，关键在于：若动作规划器缺乏具身化世界模型，仅凭大语言模型（LLM）的内部知识无法模拟物理定律。这一局限性阻碍了智能体在物理世界中准确有效地完成各类任务，导致网络空间与物理世界之间的无缝衔接无法实现。

3) *多样化环境中的可扩展性*：具身智能体的可扩展性体现在通过强大的感知能力、高效的决策机制和资源优化能力，适应规模更大、环境更复杂多样的场景需求。

Strategies include hierarchical SLAM for mapping, multi-modal perception, and energy-efficient edge computing. Collaborative scalability is enhanced via multi-agent systems and decentralized communication, while generalization relies on domain adaptation to operate in new environments.

VII. SIM-TO-REAL ADAPTATION

Sim-to-Real adaptation in embodied AI refers to the process of transferring capabilities or behaviors learned in simulated environments (cyber space) to real-world scenarios (physical world). It involves validating and improving the effectiveness of algorithms, models, and control strategies developed in simulation to ensure they perform robustly and reliably in physical environments. To achieve sim-to-real adaptation, embodied world models, data collection and training methods, and embodied control algorithms are three essential components.

A. Embodied World Model

Sim-to-Real involves creating simulation-based world models that closely resemble real-world environments. These models predict the next state to make decisions and are trained from scratch on physical world data, unlike VLA models which are pre-trained on large-scale datasets and fine-tuned with real-world data. World models are effective for structured tasks like autonomous driving and object sorting but are less suited for unstructured, complex tasks.

Learning world models is promising of the physical simulation field. Compared to traditional simulation methods, it offers significant advantages, such as the ability to reason about interactions with incomplete information, meet real-time computation requirements, and improve prediction accuracy over time. The predictive capability of such world models is crucial, enabling robots to develop the physical intuition necessary to operate in the human world. As shown in Fig. 10, according to the learning pipeline of the world environment, they can be divided into Generation-based methods, Prediction-based methods and Knowledge-driven methods.

1) *Generation-based Methods*: As the scale of models and data increases, generative models have demonstrated the ability to understand and generate images (e.g., World Models [203]), videos (e.g., Sora [16], Pandora [204]), point clouds (e.g., 3D-VLA [205]) or other formats of data (e.g., DWM [206]) that conform to physical laws. This capability suggests that generative models can internalize world knowledge. Specifically, after exposure to large datasets, these models not only capture statistical properties but also simulate physical and causal relationships through their intrinsic structures. Thus, generative models function as more than just pattern recognition tools; they exhibit characteristics of world models. The embedded world knowledge in these models can be harnessed to enhance the performance of other models. By leveraging this knowledge, we can improve model generalization, robustness, adaptability to new environments, and predictive accuracy on unseen data [204], [205]. However, generative models also have certain limitations and drawbacks. For instance, they may produce inaccurate or distorted outputs when faced with biased data distributions or insufficient training

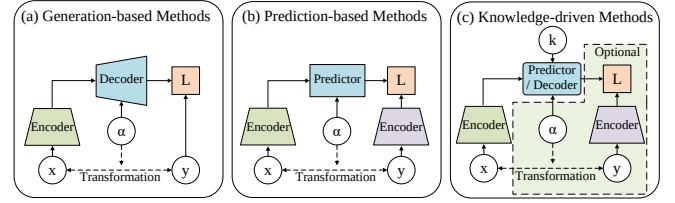


Fig. 10. Embodied world models can be roughly divided into three type. (a) **Generation-based Methods** learn the transformation relation between the input space and the output space using an autoencoder framework. (b) **Prediction-based Methods** are more general frameworks where a world model is trained in latent space. (c) **Knowledge-driven Methods** inject artificially constructed knowledge into the model, giving the model world knowledge to obtain output that meets the given knowledge constraints. Note that the components within the dashed line are optional.

data. Moreover, these models require substantial computational resources and time for training and often lack interpretability, which hinders practical application. While generative models have shown promise in generating content that adheres to physical laws, challenges such as improving efficiency, enhancing interpretability, and mitigating data bias must be addressed for broader application. Continued research is likely to unlock further value and potential in these models.

2) *Prediction-based Methods*: The prediction-based world model predicts and understands the environment by constructing and utilizing internal representations. By reconstructing corresponding features in the latent space based on provided conditions, it captures deeper semantics and associated world knowledge. This model maps input information to a latent space and operates within that space to extract and utilize high-level semantic information, thereby enabling the robots to perceive the essential representation of the world environment (e.g., I-JEPA [15], MC-JEPA [207], A-JEPA [208], Point-JEPA [209], IWM [210]) and more accurately perform embodied downstream tasks (e.g., iVideoGPT [211], IRASim [212]), STP [213], MuDreamer [214]). Latent features, unlike pixel-level information, can abstract and decouple various forms of knowledge, enabling models to handle complex tasks and scenes more effectively while enhancing generalization [215]. In spatiotemporal modeling, for instance, a world model predicts an object's post-interaction state by integrating its current state, the nature of the interaction, and internal knowledge. Specifically, embodied world models generate dynamic environmental predictions by combining perceptual information with prior knowledge, relying on both sensory data and inherent world knowledge to accurately infer and predict environmental changes [211], [213], [214]. This process considers the current state of objects alongside their historical and contextual information.

Similarly, leveraging the world knowledge embedded in its representations can further enhance the model's perception and robustness [15], [207], [210], [216]. By operating in latent space, it is expected that robots can maintain high performance in different environments at a lower cost [214]. The key to this approach lies in abstract processing and knowledge decoupling, enabling efficient adaptation to complex situations. However, such models may exhibit limitations and instability when dealing with previously unseen environments and conditions. Additionally, the world knowledge decoupled

核心策略包含分层式SLAM地图构建、多模态感知技术以及节能型边缘计算方案。通过多智能体系统与去中心化通信机制,系统可实现协同扩展能力;其泛化能力则依托领域适应机制,确保在新环境中持续稳定运行。

VII.SIM-TO-实际适应

具身人工智能中的仿真到现实适应技术,是指将模拟环境(网络空间)中习得的能力或行为迁移至真实场景(物理世界)的过程。该技术需要验证并优化仿真环境中开发的算法、模型及控制策略的有效性,确保其在物理环境中具备稳健可靠的表现。要实现仿真到现实的适应性迁移,需依托三个核心要素:具身世界模型、数据采集与训练方法,以及具身控制算法。

A. 具身世界模型

Sim-to-Real技术的核心在于构建高度模拟现实环境的仿真世界模型。这些模型能够预测系统下一状态以辅助决策,并基于物理世界数据从零开始训练;而VLA模型则是在大规模数据集上预先训练,并利用真实世界数据进行微调。此类世界模型在自动驾驶和物体分类等结构化任务中表现优异,但对于非结构化、复杂的任务则适用性较低。

学习世界模型在物理仿真领域展现出巨大潜力。相较于传统仿真方法,该技术具有显著优势:能够基于不完整信息进行交互推理、满足实时计算需求,并随时间推移提升预测精度。这类世界模型的预测能力至关重要,它能帮助机器人培养在人类环境中操作所需的物理直觉。如图10所示,根据世界环境的学习流程,相关方法可分为基于生成的方法、基于预测的方法以及知识驱动方法三大类。

1) **基于生成的方法**:随着模型和数据规模的扩大,生成模型已展现出理解和生成图像(如世界模型[203])、视频(如Sora[16]、Pandora[204])、点云(如3D-VLA[205])或其他符合物理定律的数据格式(如DWM[206])的能力。这一能力表明生成模型能够内化世界知识。具体而言,在接触大规模数据集后,这些模型不仅能捕捉统计特性,还能通过其内在结构模拟物理与因果关系。因此,生成模型不仅仅是模式识别工具,更展现出世界模型的特征。这些模型中嵌入的世界知识可被利用以提升其他模型的性能。通过利用这些知识,我们可以提高模型的泛化能力、鲁棒性、对新环境的适应性以及对未见数据的预测准确性[204]、[205]。然而,生成模型也存在某些局限性和缺陷。例如,当面对存在偏见的数据分布或训练数据不足时,这类模型可能会产生不准确或失真的输出结果。此外,这些模型在训练过程中需要消耗大量计算资源和时

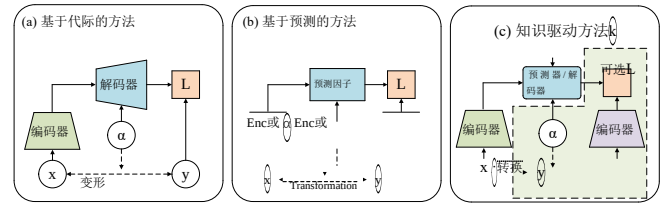


图10. 实体化世界模型大致可分为三种类型: (a)基于生成的方法通过自编码器框架学习输入空间与输出空间之间的转换关系; (b)基于预测的方法属于更通用的框架,其世界模型在潜在空间中进行训练; (c)知识驱动方法通过向模型注入人工构建的知识,使模型具备世界知识以生成符合给定知识约束的输出结果。需注意虚线框内的组件均为可选配置。

间,且往往缺乏可解释性,这严重制约了实际应用效果。尽管生成式模型在生成符合物理定律的内容方面展现出潜力,但要实现更广泛的应用,仍需解决效率提升、增强可解释性以及缓解数据偏见等关键挑战。持续的研究投入很可能会为这类模型释放更多价值与应用潜力。

2) **基于预测的方法**:基于预测的世界模型通过构建和利用内部表征来预测和理解环境。通过根据给定条件在潜在空间中重建相应特征,该模型能够捕捉更深层次的语义及相关世界知识。该模型将输入信息映射到潜在空间,并在该空间内操作以提取和利用高层次语义信息,从而使机器人能够感知世界环境的本质表征(例如I- JEPa[15]、MC- JEPa[207]、A- JEPa[208]、Point- JEPa[209]、IWM[210]),并更准确地执行具身化下游任务(例如iVideo-GPT[211]、IRASim[212]、STP[213]、MuDreamer[214])。与像素级信息不同,潜在特征能够抽象并解耦多种形式的知识,使模型在增强泛化能力的同时更有效地处理复杂任务和场景[215]。以时空建模为例,世界模型通过整合物体的当前状态、交互性质及内部知识,来预测物体在交互后的状态。具体而言,具身化世界模型通过将感知信息与先验知识相结合,依托感官数据和固有世界知识,能够准确推断并预测环境变化,从而生成动态环境预测[211]、[213]、[214]。该过程不仅考虑物体的当前状态,还会综合考量其历史数据与上下文信息。

类似地,利用其表征中嵌入的世界知识可以进一步提升模型的感知能力和鲁棒性[15]、[207]、[210]、[216]。通过在潜在空间中运行,预计机器人能够以更低的成本在不同环境中保持高性能[214]。该方法的关键在于抽象处理与知识解耦,从而能够高效适应复杂情境。然而,此类模型在处理未见过的环境和条件时可能表现出局限性和不稳定性。此外,世界知识被解耦后

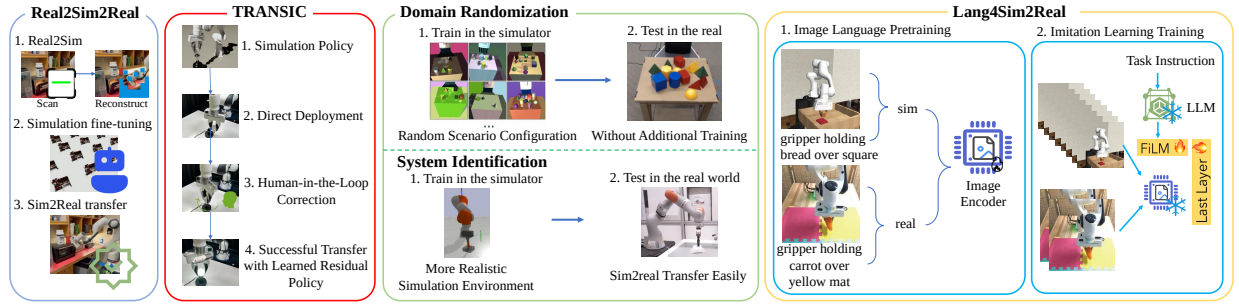


Fig. 11. Five pipelines to achieve sim-to-real gap. “**Real2Sim2Real**” reduces the gap by reconstructing real scenes. “**TRANSIC**” compensates for the sim-to-real transfer gap through human-corrected interventions. “**Domain Randomization**” enhances model transfer adaptability by simulating environmental diversity. “**System Identification**” improves sim-to-real environment similarity, thereby mitigating the sim-real gap. “**Lang4Sim2Real**” uses natural language to bridge two domains, learning invariant image representations and reducing visual gaps.

in the latent space may have interpretability issues.

3) *Knowledge-driven Methods*: Knowledge-driven world models inject artificially constructed knowledge into the models, endowing them with world knowledge. This method has shown broad application potential in the field of embodied AI. For example, in the real2sim2real approach [217], real-world knowledge is used to build physics-compliant simulators, which are then used to train robots, enhancing model robustness and generalization capabilities. Additionally, artificially constructing common sense or physics-compliant knowledge and applying them to generative models or simulators is a common strategy (e.g., ElastoGen [218], One-2-3-45 [219], PLot [220]). This approach imposes more physically accurate constraints on the model, enhancing its reliability and interpretability in generation tasks. These constraints ensure the model’s knowledge is both accurate and consistent, reducing uncertainty during training and application. Some approaches combine artificially created physical rules with LLMs or MLMs. By leveraging the commonsense capabilities of LLMs and MLMs, these approaches (e.g., Holodeck [56], LEGENT [221], GRUtopia [222]) generate diverse and semantically rich scenes through automatic spatial layout optimization. This greatly advances the development of general-purpose embodied agents by training them in novel and diverse environments.

4) *Limitations*: Current limitations of world models include handling the complexity and variability of real-world environments, such as high-dimensional sensory inputs, dynamic and stochastic elements, and long-term dependencies. Many models struggle with generalization across unseen scenarios, often requiring extensive training data and computational resources. Additionally, sim-to-real transfer remains problematic, as simulated environments fail to fully capture real-world physics and noise. These limitations can be addressed by integrating more realistic simulators, incorporating multimodal sensory inputs, and using hierarchical or modular architectures. Improving data efficiency, enhancing transfer learning techniques, and incorporating real-world priors can also enable more accurate predictions and adaptive decision-making. Recently, the Cosmos Platform [223] was proposed by NVIDIA, which contains autoregressive and diffusion models for Text-to-World and Video-to-World generation, to accelerate the development of physical AI systems. It may give us some inspirations about how to build an embodied world model.

B. Data Collection and Training

For sim-to-real adaptation, the high-quality data is important. Traditional data collection methods involve expensive equipment and precise operations, which are time-consuming and labor-intensive. Recently, some efficient and cost-effective methods have been proposed for high-quality demonstration data collection and training. We discuss data collection methods in both real-world and simulated environments.

1) *Real-World Data*: Training large, high-capacity models with rich datasets has shown great success. This approach is also promising for robotics, where large, diverse datasets can enhance generalization and adaptability. Open X-Embodiment [202] provides data from 22 robots with 527 skills and 160,266 tasks in domestic settings. UMI [224] offers a framework for collecting dynamic, bimanual data using a handheld gripper. Mobile ALOHA [225] enables data collection for full-body mobile manipulation tasks. Human-agent collaboration [226] improves data quality and efficiency by combining human input with agent refinement processes.

2) *Simulated Data*: Data collection in real-world settings is resource-intensive and time-consuming, making simulation-based data collection an attractive alternative. Simulated environments allow for automated, efficient data collection. For example, CLIPORT [168] and Transporter Networks [227] utilized Pybullet simulator data for training and successfully transferred models to real-world applications. GPartNet [228] developed a large-scale dataset with detailed part-level annotations for better object interaction in both simulations and reality. SemGrasp [167] created the CapGrasp dataset for semantically rich hand grasping in virtual environments.

3) *Sim-to-Real Paradigms*: Recently, several sim-to-real paradigms have been introduced, to mitigate the need for extensive and costly real-world demonstration data by conducting extensive learning in simulation environments, followed by migration to real-world settings. This section outlines five paradigms for sim-to-real transfer, as shown in Fig. 11.

Real2Sim2real [229] improves imitation learning by using reinforcement learning in a “digital twin” simulation to develop strategies, which are then transferred to the real world. TRANSIC [230] reduces the sim-to-real gap through real-time human intervention and residual policy training based on corrected behaviors. Domain Randomization [231]–[233] increases model generalization by varying simulation parameters to cover real-world conditions. System Identifica-

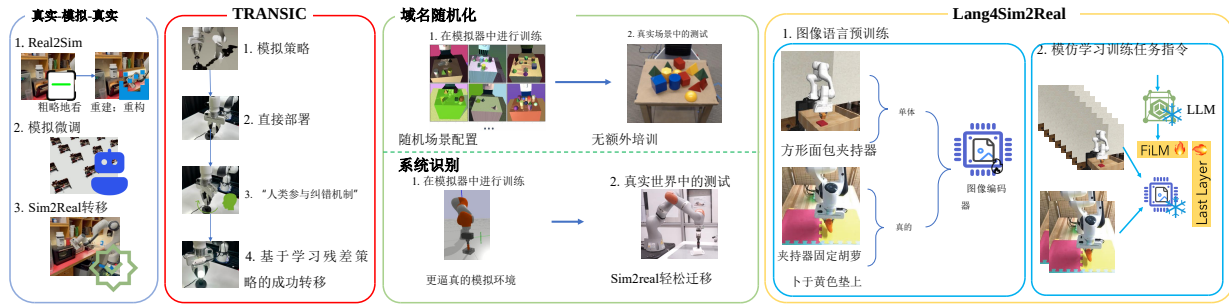


图11. 实现模拟-真实差距的五条路径。**Real2Sim2Real**通过重建真实场景来缩小差距；**transic**通过人工校正干预补偿模拟-真实迁移差距；**Domain Randomization**通过模拟环境多样性提升模型迁移适应性；**System Identification**增强模拟与真实环境的相似性，从而缓解模拟-真实差距；**Lang4Sim2Real**利用自然语言连接两个领域，学习不变的图像表示并减少视觉差距。

在潜在空间中可能存在可解释性问题。

3) **知识驱动方法**: 知识驱动的世界模型将人工构建的知识注入模型中，赋予其世界知识。该方法在具身人工智能领域展现出广泛的应用潜力。例如，在**real2sim2real**方法[217]中，利用现实世界知识构建符合物理规律的模拟器，进而用于机器人训练，从而提升模型的鲁棒性和泛化能力。此外，人工构建常识或符合物理规律的知识并将其应用于生成模型或模拟器是一种常见策略（如**ElastoGen** [218]、**One-2-3-45** [219]、**PLoT** [220]）。该方法对模型施加更精确的物理约束，提升其在生成任务中的可靠性与可解释性。这些约束确保模型知识既准确又一致，降低训练与应用过程中的不确定性。部分方法将人工创建的物理规则与大语言模型（LLMs）或多语言模型（MLMs）相结合。通过利用LLMs与MLMs的常识能力，这些方法（例如**Holodeck** [56]、**LEGENT** [221]、**GRUtopia** [222]）通过自动空间布局优化生成多样化且语义丰富的场景。这通过在新颖多样的环境中训练通用具身智能体，极大地推动了其发展进程。

4) **局限性**: 当前世界模型的主要局限在于难以处理真实环境的复杂性和多样性特征，例如高维感官输入、动态随机要素及长期依赖关系。多数模型在跨场景泛化能力方面存在短板，通常需要大量训练数据和计算资源支持。此外，模拟到现实的迁移学习仍面临挑战，因为模拟环境难以完全复现真实世界的物理规律与噪声特征。这些局限可通过集成更逼真的仿真器、整合多模态感官输入以及采用分层或模块化架构来解决。提升数据利用效率、优化迁移学习技术并融入现实世界先验知识，也有助于实现更精准的预测和自适应决策。近期，NVIDIA提出的**Cosmos**平台[223]通过为文本生成世界模型和视频生成世界模型配备自回归模型与扩散模型，加速了物理AI系统的研发进程。该平台为构建具身化世界模型提供了重要参考范例。

B. 数据收集与培训

在仿真到真实场景的适配过程中，高质量数据至关重要。传统数据采集方法需要昂贵的设备和精密的操作流程，既耗时又劳动密集。近年来，已提出多种高效且成本效益高的方法，可用于高质量演示数据的采集与训练。本文将探讨现实环境与模拟环境中数据采集的方法。

1) **真实世界数据**: 利用丰富数据集训练大型高容量模型已取得显著成效。该方法在机器人领域同样展现出广阔前景，海量多样化数据集能有效提升模型的泛化能力和适应性。**OpenX-Embodiment** [202]平台提供了包含22款机器人、527项技能及160,266个家庭场景任务的数据集。**UMI** [224]通过手持抓取器构建了动态双臂操作数据采集框架。**MobileAloha** [225]系统实现了全身移动操作任务的数据采集。人机协作[226]通过整合人工输入与智能体优化流程，显著提升了数据质量与采集效率。

2) **模拟数据**: 在现实场景中进行数据收集既耗费资源又耗时，因此基于模拟的数据收集成为一种颇具吸引力的替代方案。模拟环境能够实现自动化、高效的数据收集。例如，**cliptor** [168]和**Transporter Networks** [227]利用**Pybullet**模拟器数据进行训练，并成功将模型迁移至现实应用中；**GAPartNet** [228]构建了包含详细部件级标注的大规模数据集，以提升模拟与现实场景中的物体交互精度；**SemGrasp** [167]则创建了**CapGrasp**数据集，用于虚拟环境中语义丰富的抓取任务。

3) **模拟到真实范式**: 近年来，为减少对大量昂贵真实世界演示数据的需求，学界提出了多种模拟到真实范式。这些范式通过在仿真环境中进行大规模学习，再迁移至实际应用场景来实现目标。如图11所示，本节将阐述五种模拟到真实迁移范式。

Real2Sim2real [229]通过在“数字孪生”模拟环境中运用强化学习来改进模仿学习，从而开发策略并将其迁移至现实世界。**transic** [230]通过实时人工干预及基于修正行为的残差策略训练，缩小模拟与现实之间的差距。**Domain Randomization** [231]–[233]通过调整模拟参数以覆盖现实条件，提升模型泛化能力。**System Identifica-**

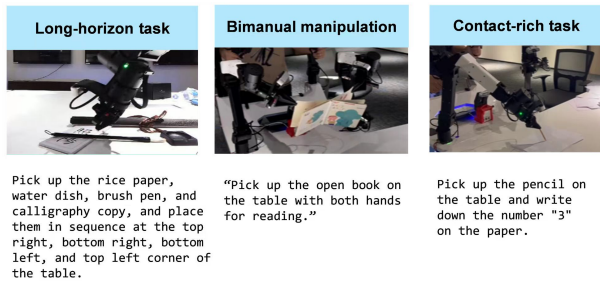


Fig. 12. Exemplar tasks from ARIO, where the top row indicates the task category while the text at the bottom row provides task instructions.

tion [234], [235] creates accurate simulations of real-world scenes to ensure smooth transitions from simulation to reality. Lang4sim2real [236] leverages natural language descriptions to bridge the sim-to-real gap, improving model generalization with cross-domain image representations.

4) *ARIO (All Robots in One)*: Despite the seemingly unified structure of pre-training datasets like Open X-embodiment, several critical issues remain unresolved. These issues include the absence of comprehensive sensory modalities—no dataset currently integrates images, 3D vision, text, tactile, and auditory inputs simultaneously. Additionally, the lack of a unified format in multi-robot datasets complicates data processing and loading. Furthermore, there is an incompatibility in representing diverse control objects across different robotic platforms, insufficient data volume that hinders large-scale pretraining, and a scarcity of datasets that combine both simulated and real data, which is essential for addressing the sim-to-real gap.

To overcome these challenges, we propose ARIO (All Robots In One), which is a new dataset standard that optimizes existing datasets and facilitates the development of more versatile and general-purpose embodied agents. The ARIO standard [237] records control and motion data from robots with different morphologies in a unified format. ARIO’s unified format accommodates variable data from diverse robot types, ensuring precise timestamps. This standard enables users to efficiently train high-performing, generalizable embodied AI models, positioning ARIO as the ideal format for embodied AI datasets. Building on the ARIO standard, a unified large-scale ARIO dataset is further developed, which comprises approximately 3 million episodes collected from 258 series and 321,064 tasks. Fig. 12 shows exemplar tasks from ARIO.

The ARIO dataset addresses the limitations of current datasets and facilitates the development of robust, general-purpose embodied agents. By providing a cohesive framework for data collection and representation, ARIO paves the way for the development of more powerful and versatile embodied agents, capable of navigating and interacting with the physical world in complex and diverse ways.

5) *Real-world Deployments of Embodied AI Systems*: Embodied AI systems have made significant strides across various scenes. In healthcare, robots like the Da Vinci Surgical System and Moxi automate tasks such as surgery precision and supply deliveries. In logistics, Amazon Robotics and Boston Dynamics’ Stretch improve efficiency in warehousing and transportation. Manufacturing benefits from AI-driven robots like Fanuc and ABB, enhancing precision and collaboration.

Nevertheless, sim-to-real adaptation faces significant challenges, including the domain gap between simulation and real-world data distributions, the complexity of dynamic real-world interactions, and the limited diversity of training data. Models often overfit to simulations, struggle with real-world sensor noise, and fail to handle unexpected events.

VIII. CHALLENGES AND FUTURE DIRECTIONS

Despite of the rapid progress of embodied AI, it faces several challenges and presents exciting future directions.

High-quality Robotic Datasets: Obtaining sufficient real world robotic data remains a significant challenge. Collecting this data is both time-consuming and resource-intensive. Relying solely on simulation data worsens the sim-to-real gap problem. Creating diverse real world robotic datasets necessitates close and extensive collaboration among various institutions. Additionally, the development of more realistic and efficient simulators is essential for improving the quality of simulated data. For building generalizable embodied models capable of cross-scenario and cross-task applications in robotics, it is essential to construct large-scale datasets, leveraging high-quality simulated environment data to assist real world data.

Long-Horizon Task Execution: Executing single instructions can often entail long-horizon tasks for robots, exemplified by commands like “clean the kitchen”, which involve activities such as rearranging objects, sweeping floors, wiping tables, and more. Accomplishing such tasks successfully necessitates the robot’s ability to plan and execute a sequence of low-level actions over extended time spans. While current high-level task planners have shown initial success, they often prove inadequate in diverse scenarios due to their lack of tuning for embodied tasks. Addressing this challenge requires the development of efficient planners equipped with robust perception capabilities and much commonsense knowledge. To balance the trade-off between planning complexity and real-time adaptability, we can combine lightweight monitor modules for high-frequency monitoring, and two adapters for subtask and path adaptation reasoning at a lower frequency.

Causal Reasoning: Existing data-driven embodied agents make decisions based on intrinsic data correlations. However, this approach does not allow the models to truly understand the causal relations between knowledge, behavior, and environment, resulting in biased strategies. This makes it difficult to ensure that they can operate in real-world environments in a robust and reliable manner. Therefore, it is important for embodied agents to be driven by world knowledge, capable of autonomous causal reasoning. By understanding the world through interaction and learning its workings via abductive reasoning, we can further enhance the adaptability, decision reliability, and generalization capabilities of embodied agents in complex real-world environments. For embodied tasks, it is necessary to establish spatial-temporal causal relations across modalities through interactive instructions and state predictions. Moreover, agents need to understand the affordances of objects to achieve adaptive task planning in dynamic scenes.

Unified Evaluation Benchmark: While numerous benchmarks exist for evaluating low-level control policies, they vary

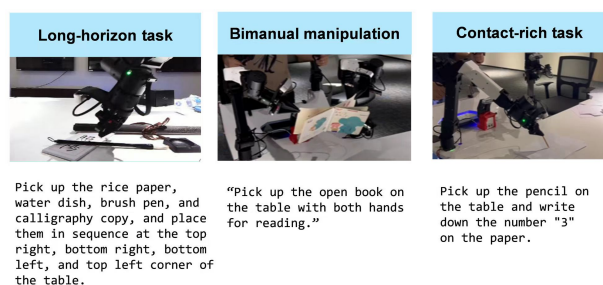


图12.来自 ARIO 的示例任务，其中第一行表示任务类别，而底部行的文字提供任务说明。

tion[234]、[235]能够精准模拟真实场景，确保从模拟到现实的过渡过程流畅自然。Lang4sim2real[236]通过自然语言描述弥合模拟与现实之间的差距，借助跨域图像表征提升模型泛化能力。

4) *ARIO (All Robots in One)*: 尽管像 Open X-embodiment 这样的预训练数据集在结构上看似统一，但仍存在若干关键问题尚未解决。这些问题包括缺乏全面的感知模态——目前尚无数据集能同时整合图像、三维视觉、文本、触觉和听觉输入。此外，多机器人数据集缺乏统一格式，导致数据处理和加载复杂化。更重要的是，不同机器人平台在表示多样化控制对象时存在兼容性问题；数据量不足阻碍了大规模预训练；而同时结合模拟数据与真实数据的数据集极为稀缺，而这正是弥合模拟与真实环境差距的关键。

为应对这些挑战，我们提出了 ARIO (All Robots In One) 标准——这一新型数据集标准通过优化现有数据集，助力开发更具通用性和多功能性的具身智能体。ARIO 标准[237]采用统一格式记录不同形态机器人的控制指令与运动数据。其标准化格式可适配各类机器人数据的多样性特征，确保时间戳信息的精确性。该标准使用户能够高效训练高性能、可泛化的具身AI模型，使 ARIO 成为具身AI数据集的理想格式。基于 ARIO 标准，我们进一步开发了统一的大规模 ARIO 数据集，包含从258个系列任务中采集的约300万次动作序列（共321,064个任务）。图12展示了 ARIO 中的典型任务示例。

ARIO 数据集克服了现有数据集的局限性，有助于开发出稳健且通用的具身智能体。通过提供一个统一的数据收集与表示框架，ARIO 为开发更强大、更灵活的具身智能体奠定了基础——这些智能体能够以复杂多样的方式在物理世界中导航并进行交互。

5) *具身人工智能系统的实际应用*: 具身人工智能系统已在多个领域取得显著进展。在医疗健康领域，达芬奇手术系统和Moxi机器人实现了手术精准度与物资配送等任务的自动化。物流领域中，亚马逊机器人系统与波士顿动力公司的Stretch机器人显著提升了仓储运输效率。制造业借助发那科和ABB等人工智能驱动机器人，实现了生产精度与协作效率的双重提升。然而，从模拟环境到真实场景的适

配仍面临重大挑战：包括仿真数据与真实数据分布的领域差异、动态现实交互的复杂性，以及训练数据多样性不足等问题。现有模型常出现过拟合现象，难以应对真实场景中的传感器噪声干扰，且对突发事件处理能力存在局限。

第八节.挑战与未来方向

尽管具身人工智能取得了快速进展，但它仍面临诸多挑战，并展现出令人振奋的未来发展方向。

高质量机器人数据集: 获取充足的现实世界机器人数据仍面临重大挑战。数据采集过程既耗时又资源密集，单纯依赖仿真数据会加剧仿真与现实之间的差距问题。构建多样化的现实世界机器人数据集需要各机构开展紧密协作。

此外，开发更逼真高效的仿真器对提升仿真数据质量至关重要。为构建具备跨场景、跨任务应用能力的通用化具身模型，必须依托高质量仿真环境数据来辅助真实世界数据，从而构建大规模数据集。**长时程任务执行**: 执行单条指令往往涉及机器人需要完成的长时程任务，例如“打扫厨房”这类指令包含重新摆放物品、拖地、擦拭桌面等操作。成功完成此类任务要求机器人具备在长时间跨度内规划并执行低级动作序列的能力。尽管当前的高级任务规划器已取得初步成功，但由于缺乏对具身任务的专门调优，它们在多种场景下往往表现欠佳。要解决这一挑战，需要开发具备强大感知能力和丰富常识知识的高效规划器。为平衡规划复杂度与实时适应性之间的权衡，我们可以结合轻量级监控模块实现高频监测，并采用两个适配器在较低频率下进行子任务与路径适应推理。**因果推理**: 现有的数据驱动型具身智能体基于数据内在关联性做出决策，但这种方法无法让模型真正理解知识、行为与环境之间的因果关系，从而导致策略存在偏差。这使得难以确保其在现实环境中能够稳健可靠地运行。因此，具身智能体必须以世界知识为驱动，具备自主因果推理能力。通过互动理解世界并运用溯因推理学习其运作机制，我们能够进一步提升具身智能体在复杂现实环境中的适应性、决策可靠性及泛化能力。针对具身任务，需要借助交互式指令和状态预测来建立跨模态时空因果关系。此外，智能体还需理解物体的可用性特征，才能在动态场景中实现自适应任务规划。

统一评估基准: 尽管现有大量用于评估低级控制策略的基准测试，但这些标准存在显著差异

significantly in the assessed skills. Furthermore, the objects and scenes in these benchmarks are typically limited by simulator constraints. To comprehensively evaluate embodied models, the benchmark should encompass a diverse range of skills using realistic simulators. Many benchmarks for high-level task planners focus on assessing planning capability through question-answering tasks. However, a more desirable approach involves evaluating both the high-level task planner and the low-level control policy together for executing long-horizon tasks and measuring success rates, rather than relying solely on isolated assessments of the planner.

Security and Privacy: Embodied agents face significant security challenges when deployed in sensitive or private spaces. These agents often rely on LLMs for decision-making, which introduces new vulnerabilities. For instance, LLMs are susceptible to backdoor attacks like word injection, scenario manipulation, and knowledge injection, which can lead to dangerous outcomes such as autonomous vehicles accelerating towards obstacles or robots performing hazardous actions. To mitigate these risks, we can evaluate potential attack vectors and develop more robust defenses. Additionally, the secure prompting, state management, and safety validation mechanisms can be used to enhance the security and robustness.

IX. CONCLUSION

Embodied AI allows agents to sense, perceive, and interact with various objects from both cyber space and physical world, which exhibits its vital significance toward achieving AGI. This survey extensively reviews embodied robots, simulators, four representative embodied tasks: visual active perception, embodied interaction, embodied agents and sim-to-real adaptation, and future research directions. The comparative summary of the embodied robots, simulators, datasets, and approaches provides a clear picture of the recent development in embodied AI, which greatly benefits the future research along this emerging and promising research direction.

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在评估技能方面存在显著差异。此外，这些基准测试中的物体和场景通常受限于模拟器的约束条件。要全面评估具身模型，基准测试应通过真实模拟器涵盖多样化技能。许多针对高级任务规划器的基准测试侧重于通过问答任务评估规划能力。然而，更理想的方法是将高级任务规划器与低级控制策略相结合进行评估，以执行长期任务并衡量成功率，而非仅依赖对规划器的孤立性评估。

安全与隐私：当嵌入式智能体被部署在敏感或私密环境中时，会面临严峻的安全挑战。这些智能体通常依赖大语言模型（LLM）进行决策，这带来了新的安全隐患。例如，LLM容易遭受词汇注入、场景操纵和知识注入等后门攻击，可能导致自动驾驶车辆冲向障碍物或机器人执行危险操作等危险后果。为降低这些风险，我们可以评估潜在的攻击途径并开发更强大的防御机制。此外，通过采用安全的提示机制、状态管理机制和安全验证机制，可进一步提升系统的安全性和鲁棒性。

IX. 结论

具身人工智能使智能体能够感知、识别并与网络空间及物理世界中的各类物体进行交互，这对实现通用人工智能（AGI）具有关键意义。本文系统梳理了具身机器人、仿真器、四大代表性具身任务（视觉主动感知、具身交互、具身智能体及仿真-现实适配）及未来研究方向。通过对具身机器人、仿真器、数据集及方法论的对比分析，清晰展现了具身人工智能领域的最新发展态势，为这一新兴且前景广阔的研究方向提供了重要参考依据。

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